

NxM Templates for SuperCDMS SNOLAB Run 4 — Ge Activation Data

K-line event selection · free-pretrigger two-exponential fits · 1x1 and NxM (PCA) templates

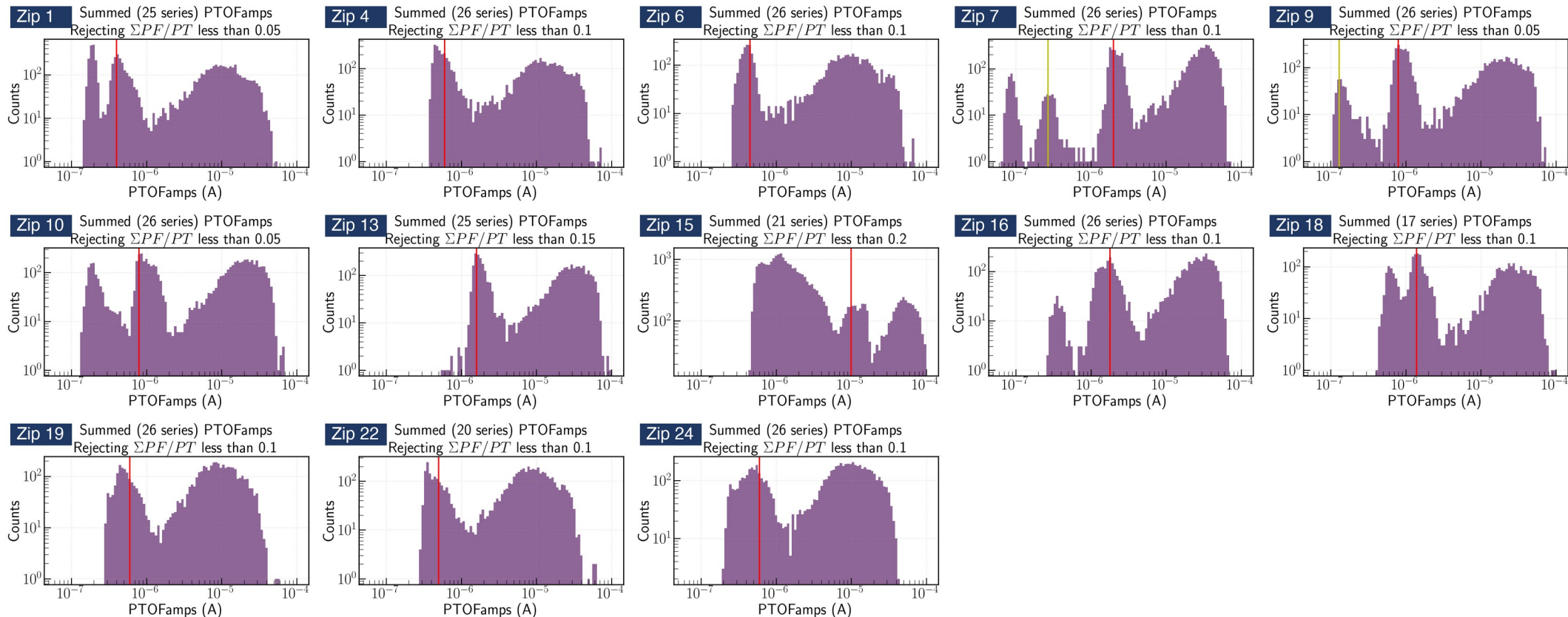
Zhiheng Li — July 2026

From the Ge-activation K-line to an event sample

Data: Ge Activation Data — Ops Shift 260612
Credit: Prof. Tarek Saab · SuperCDMS Confluence

Goal: one clean, minimally-biased pulse population per detector × channel

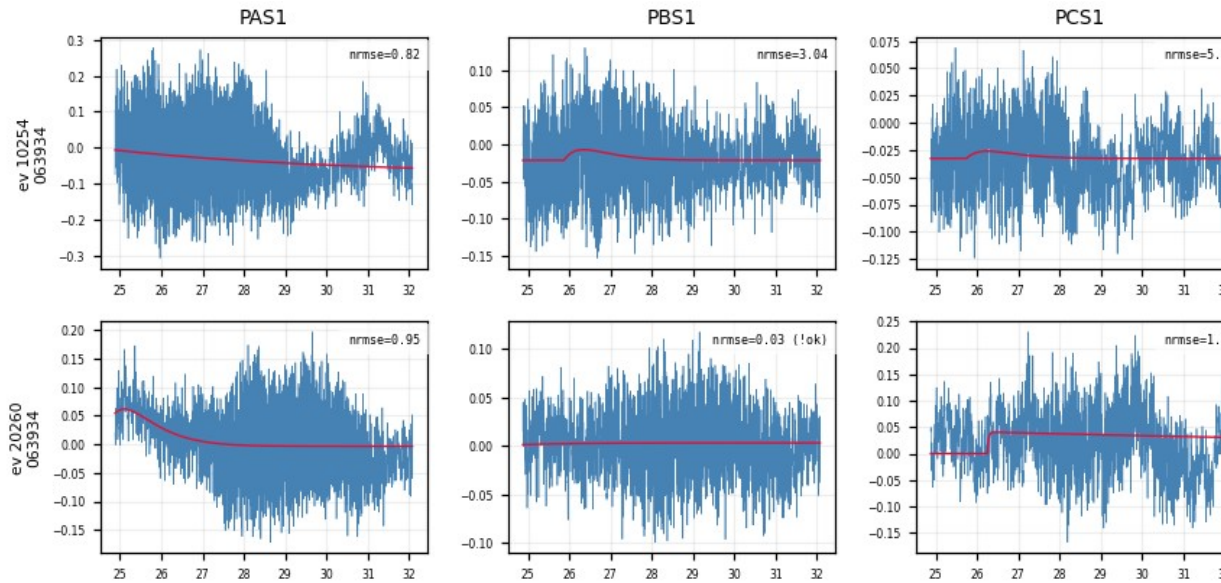
- Per detector, Prof. Saab's study marked the 10.37 keV K-line position in PTOFamps — the red line in each panel below
- Our event selection: a PTOFamps window bracketing the red line — position $\times/\div 1.35$, a rough, somewhat arbitrary eyeballed choice; deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata
- 13 detectors (zips) × 27–30 series — full raw traces cached



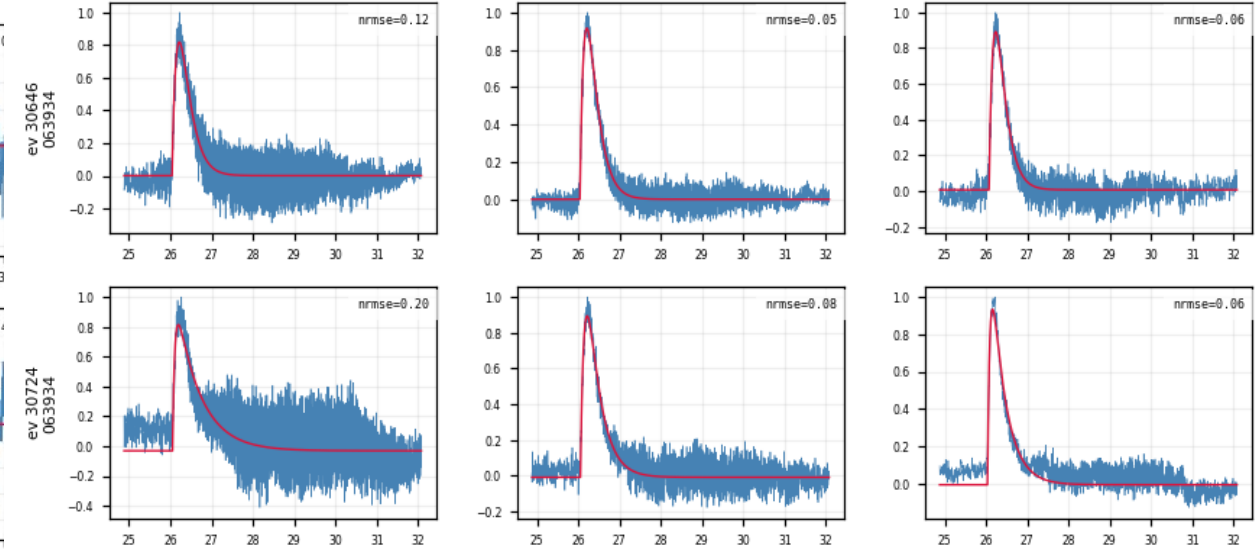
All 13 detectors, PTOFamps spectra (Prof. Saab's study)

Fit quality at a glance — event × channel grids

- One row per event, one column per channel: the low-passed trace (blue) with the fitted pulse model (red) overlaid
- A real event fits well in every channel at once; a noise trigger fails in every channel — a clean handle on which events are real



Noise triggers (Z7): the fit fails in every channel at once



K-line events (Z7): consistent good fits across channels

Per-trace algorithm: low-pass → fit → align

1. **Low-pass** — 100 kHz Butterworth
2. **Normalize** — subtract the baseline, scale the trace to unit peak
3. **Fit** — two-exponential model, all 5 parameters free (including the pretrigger t_0):

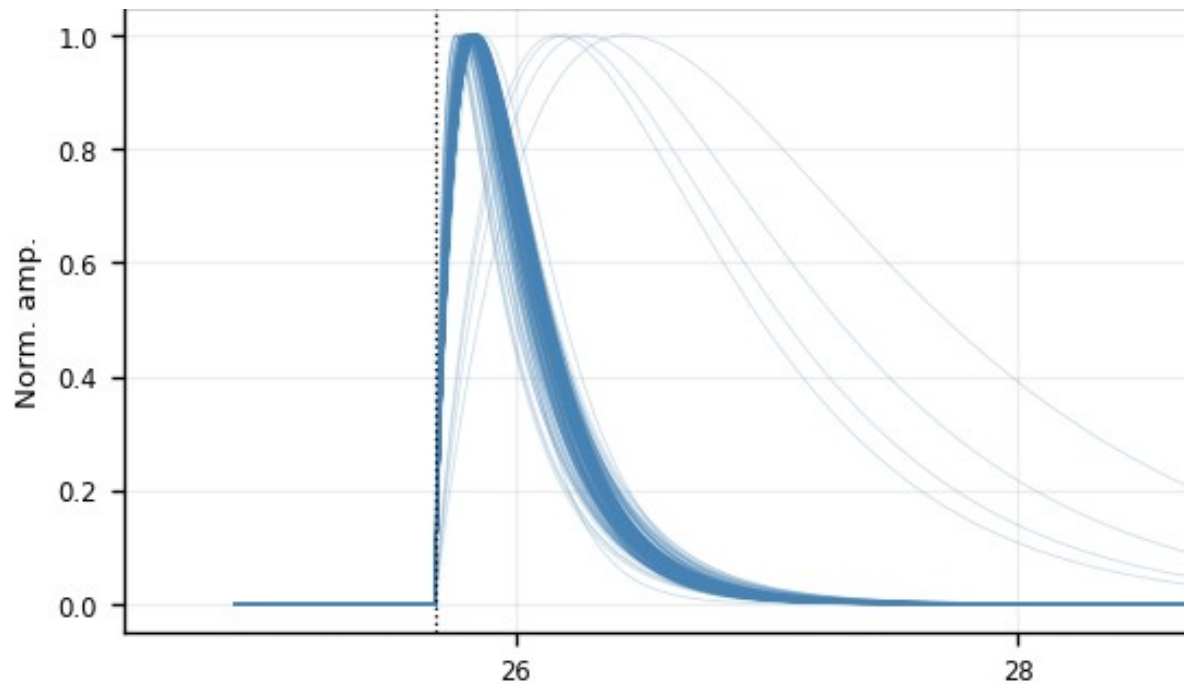
$$y(t) = A[e^{-(t - t_0)/\tau_{\text{fall}}} - e^{-(t - t_0)/\tau_{\text{rise}}}] + b$$

t_0 = pretrigger, free within 16050 ± 3000 samples (τ_{rise} , τ_{fall} : rise / fall time)

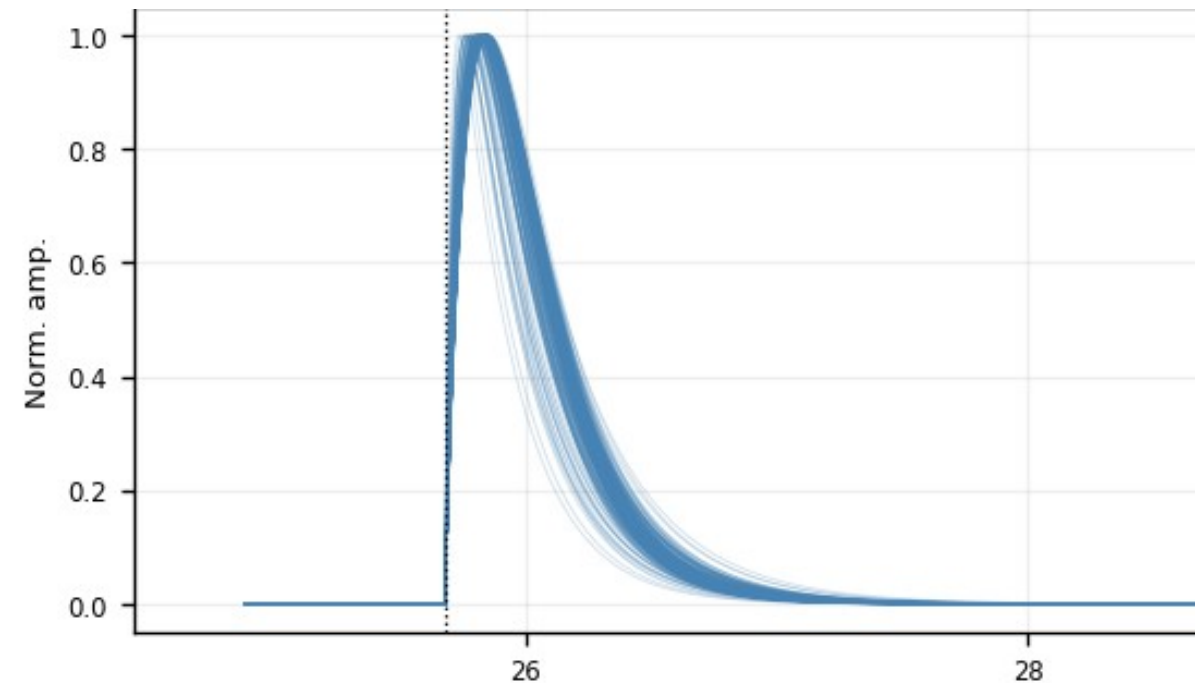
4. **Two quality checks per trace** (recorded here, not cut yet):
 - **fit_ok** — amplitude is positive, and the pulse rises faster than it falls
 - **NRMSE** — RMS of the fit residual ÷ pulse peak (smaller = better fit)

5. Align — shift each trace to a common pretrigger

- Shift the measured trace by (fitted pretrigger – 16050) — a pure translation. Drawn at the common pretrigger, the fitted curves stack into one shape family:



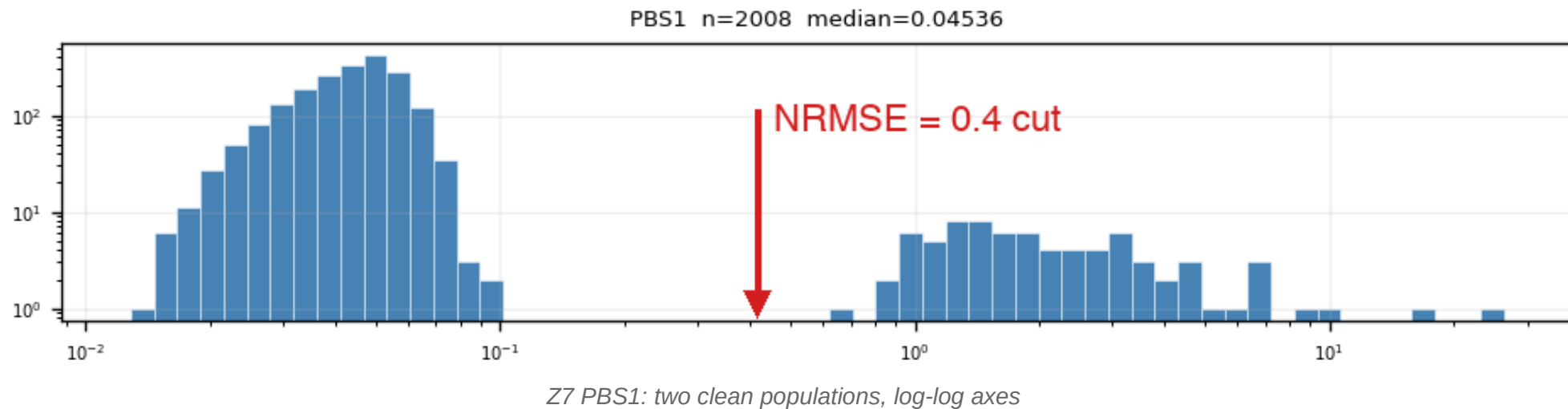
All fit_ok fitted curves (Z7 PBS1) — no cut



After $\text{NRMSE} \leq 0.4$ (Z7 PBS1) — one tight shape family

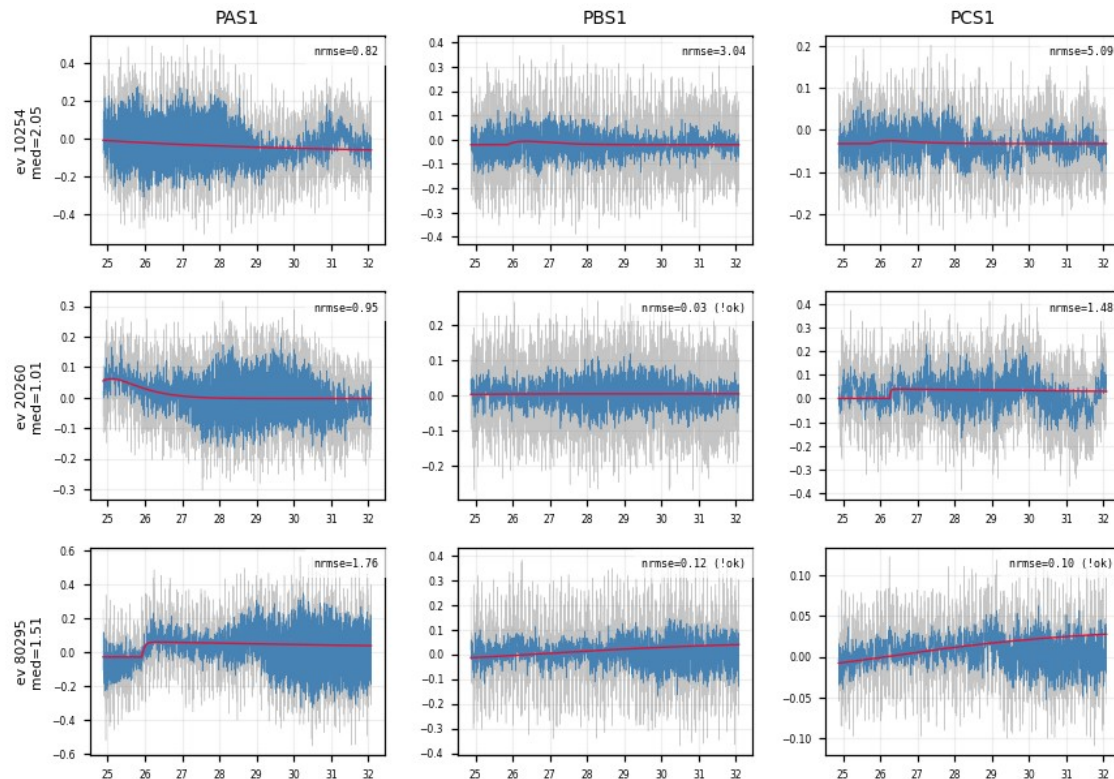
Quality cut 1: $\text{NRMSE} \leq 0.4$ — where the number comes from

- NRMSE of fit_ok events is bimodal: good fits (median ≈ 0.05 – 0.1) vs noise triggers (≈ 1 – 2), valley at ≈ 0.4 – 0.5
- The cut sits in the valley — it is read off the distribution, not tuned on the templates
- Weak detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): same bimodal picture, noise peak dominates

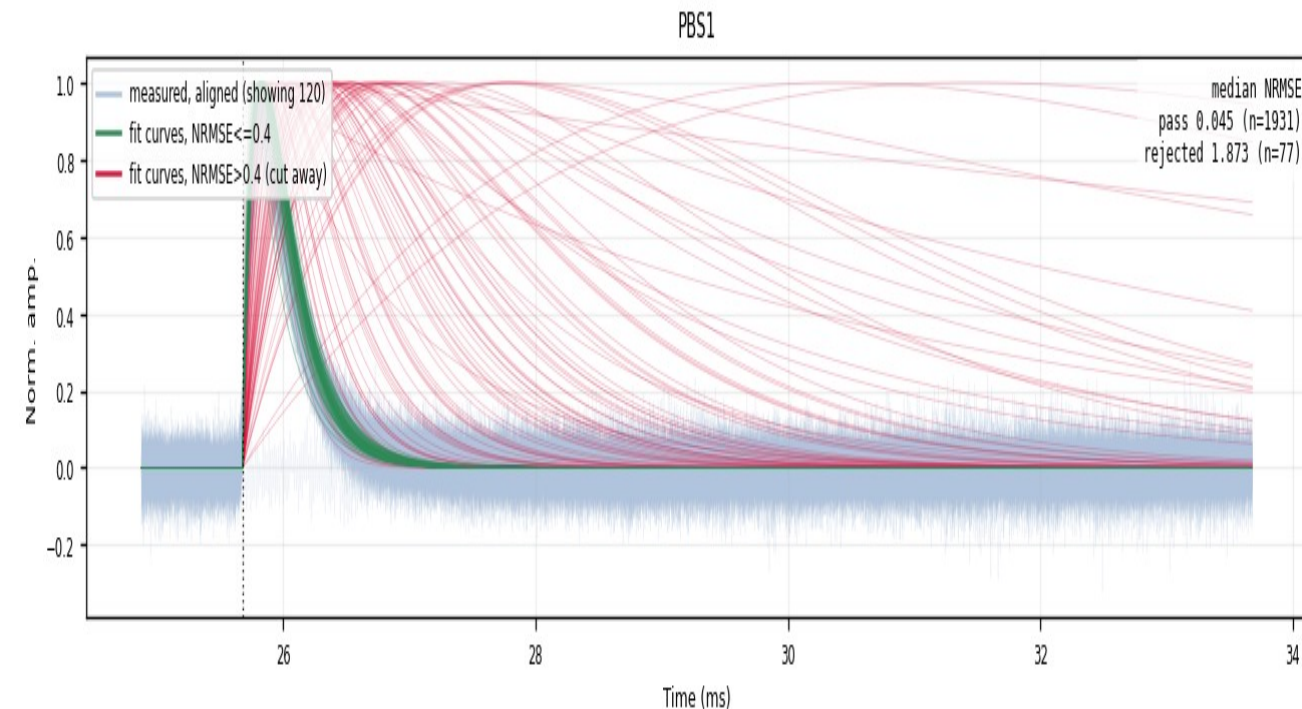


The rejected population is noise — verified in the raw traces

- Event grids of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse — the cut removes noise triggers, not physics



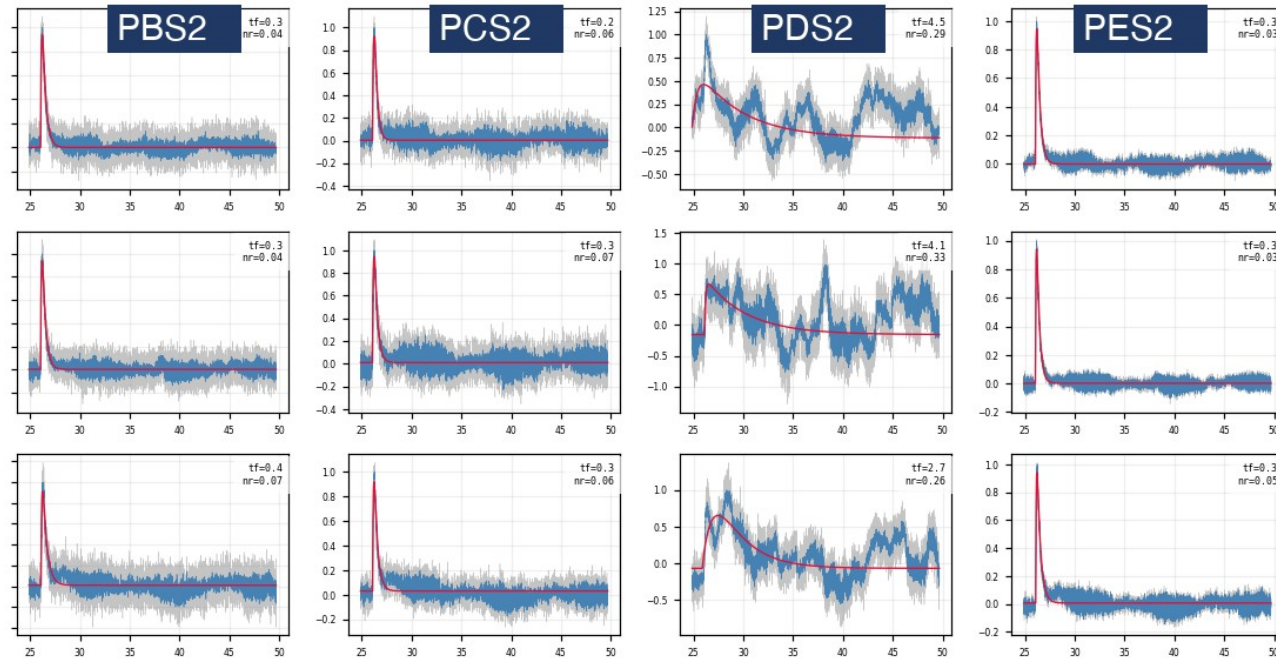
Z7: NRMSE-rejected events — raw traces are noise



Z7 PBS1 — passes the cut (kept) — cut away = rejected
faint blue = measured traces 1931 kept / 77 cut (~3.8%)

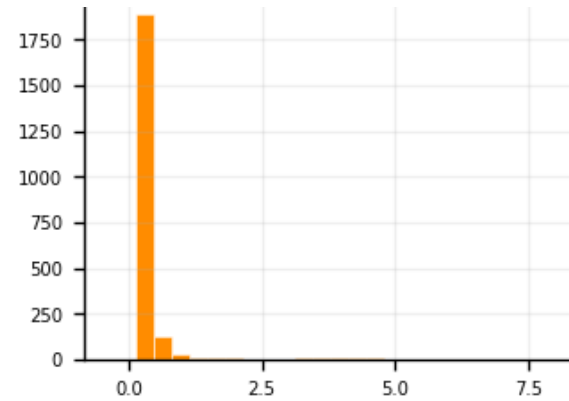
Follow-up 1 — the slow-fall tail is a one-channel artifact

- The post-cut fan still shows slow-fall tails → sample them: $\text{NRMSE} \leq 0.4$ AND $\tau_{\text{fall}} > 1.5$ ms, 10 random events, raw vs fit drawn in all 12 channels
- The sampled events are real pulses → kept. Their extreme fall times trace to one channel: only PDS2 swings (τ_{fall} median 0.51 ms vs ≈ 0.25 ms elsewhere) — a low-frequency disturbance
- Conclusion: no τ_{fall} cut — slow-fall events passing NRMSE stay in; the extreme tail is a one-channel artifact, not a slow pulse

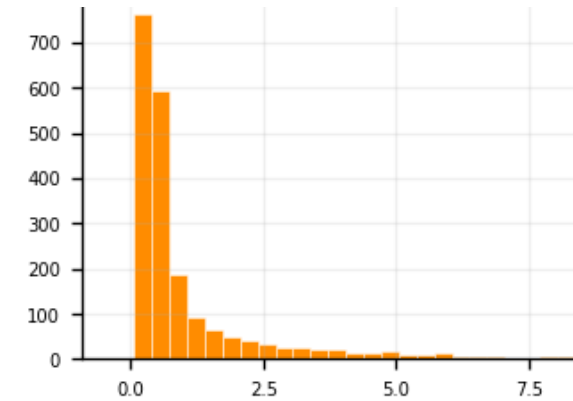


Z7, 3 sampled slow-fall events: normal in PBS2/PCS2/PES2, swings only in PDS2

Fitted τ_{fall} distribution, same 0–7 ms axis:



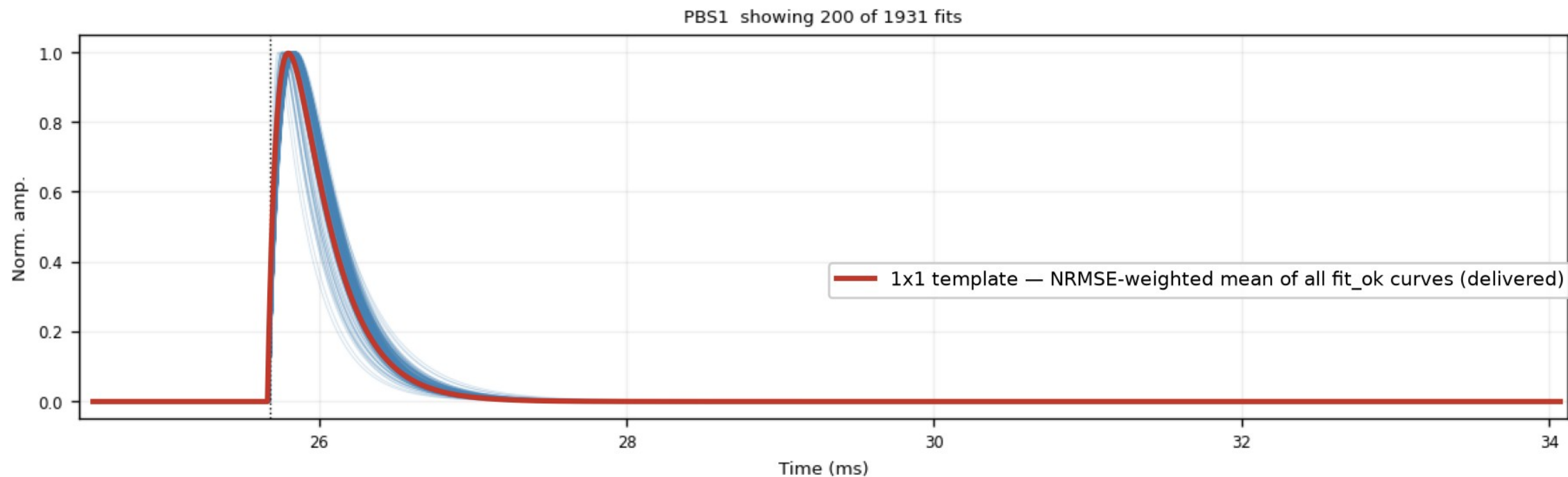
Z7 PAS1 — normal channel: narrow
(median 0.25 ms)



Z7 PDS2 — bad channel: broad tail
(median 0.51 ms)

Template family 1 — analytic 2-exp, NRMSE-weighted (1x1)

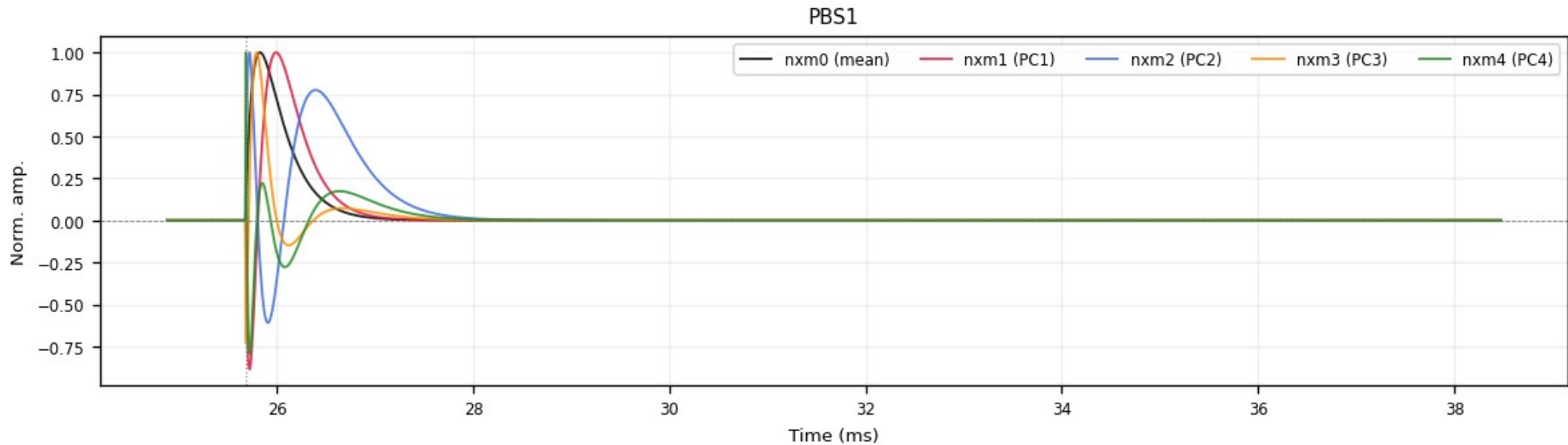
- NRMSE-weighted mean of the fit_ok 2-exp curves
- Smooth & noise-free by construction
- ROOT histograms, peak-normalized (+ summed PT / PS1 / PS2)



Z7 PBS1 — blue: the fitted curves (200 of 1931 drawn); red: the delivered 1x1 template, the NRMSE-weighted mean of all fit_ok curves, read back from the ROOT file

Template family 2 — NxM PCA templates

- Per-channel PCA over the clean fitted curves ($\text{fit_ok} + \text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3 \text{ ms}$)
- nxm0 = mean shape; nxm1 – 4 = principal components; $\text{real pulse} = \sum_i \text{amp}_i \cdot \text{nxm}_i$
- $\text{PC1} + \text{PC2} \approx 96\text{--}98\%$ of the shape variance; delivered peak-normalized



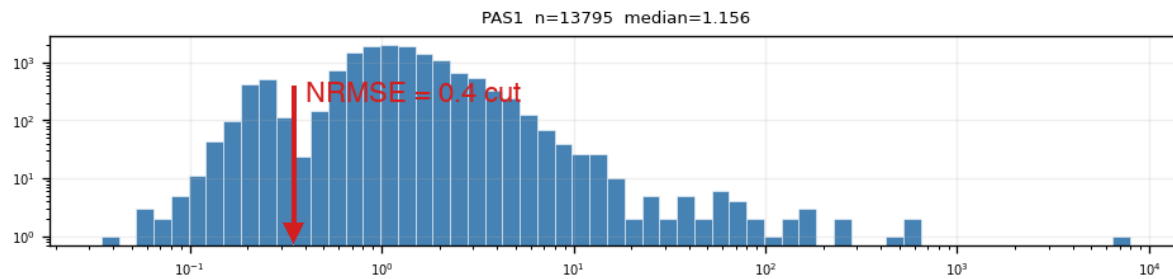
Z7 PBS1: nxm0 (mean) + nxm1 – 4 (PCs) — the oscillating components encode rise/fall-time variation

Delivered — and what remains

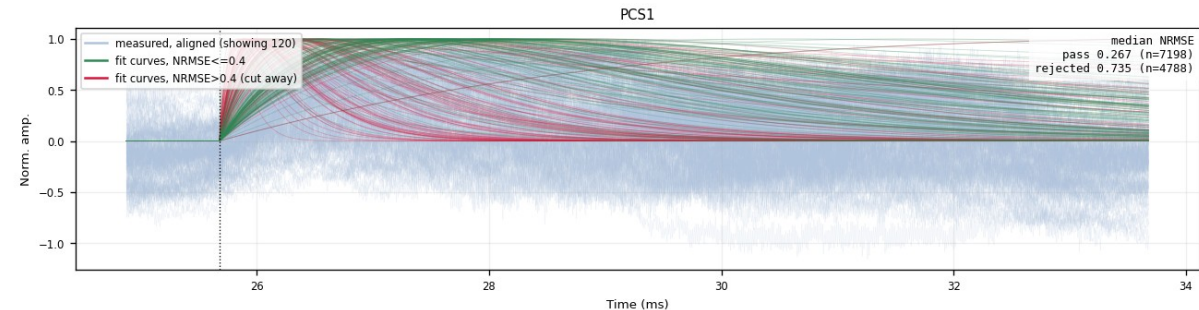
- Two template families built for all 13 detectors: 1x1 (2-exp weighted) + NxM PCA
- Delivered in the official cdmsbats PulseTemplates format
- Cuts read off the data, verified in raw traces: $\text{NRMSE} \leq 0.4$, $\tau_{\text{rise}} \leq 0.3$ ms

Backup — weak detectors (example: Z22)

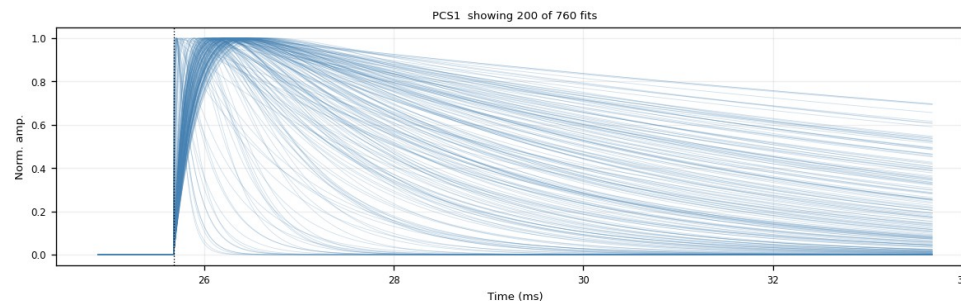
- Z1 / Z4 / Z6 / Z18 / Z19 / Z22 / Z24: K-line sits inside the noise population — the PTOF window admits a noise-dominated mixture
- Same bimodal NRMSE picture, noise peak dominates; the same 0.4 cut digs the real pulses out — Z22 PCS1: 7198 kept / 4788 cut (~40%)
- Kept bundle broader than on quiet detectors; steep red curves = fits latching onto sharp noise spikes, not fast pulses being cut
- $\tau_{\text{rise}} \leq 0.3$ ms (after NRMSE): removes 55–74% on weak zips (Z22: 71%) vs 1.6% on Z7 — residual slow drift



Z22 PAS1: NRMSE histogram — noise dominates



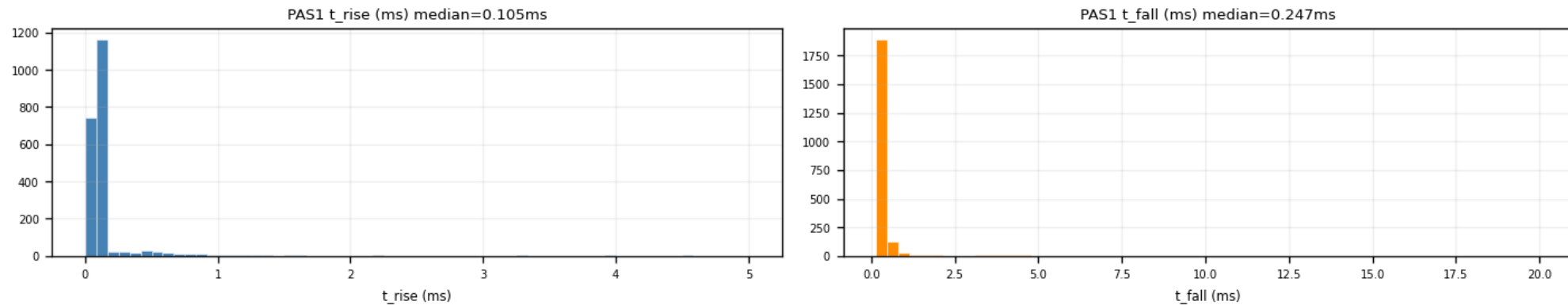
Z22 PCS1: kept (green) vs cut (red)



Z22 PCS1 after $\text{NRMSE} \leq 0.4 + \tau_{\text{rise}} \leq 0.3$ ms — final shape family (760 fits)

Backup — the $\tau_{\text{rise}} \leq 0.3$ ms ceiling (PCA input)

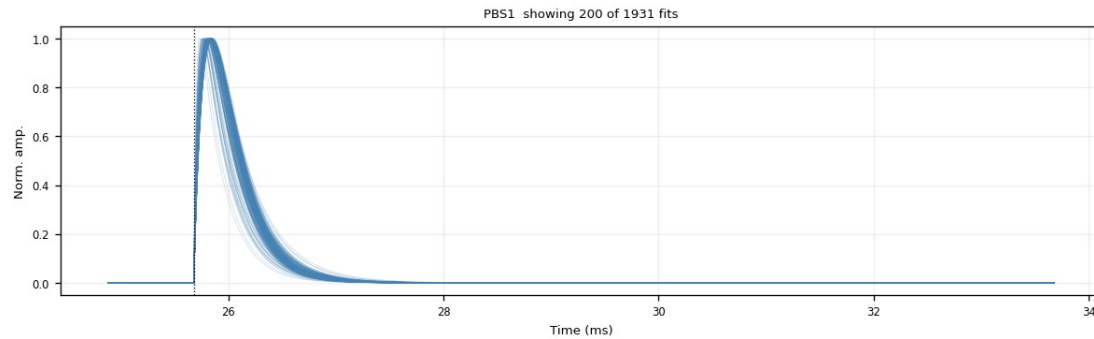
- Slow baseline drift survives NRMSE (a slow 2-exp hugs it) → mimics a slow rise
- Fast pulses at $\tau_{\text{rise}} \approx 0.1$ ms (p90 ≈ 0.15 ms) → ceiling at 0.3 ms blocks the drift tail
- Removed after the NRMSE cut: Z7 1.6% (quiet 2–6%) — weak 55–74% (Z22 71%) — all zips 54%
- Trade-off: trims the very slowest genuine pulses → decision pending



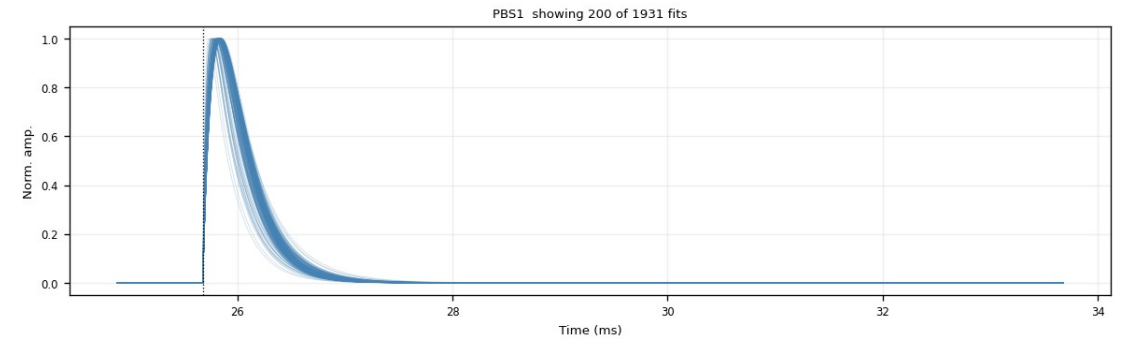
Z7 PAS1: fitted τ_{rise} (median 0.105 ms) and τ_{fall} distributions

Backup — τ_{rise} cut: a good vs a bad channel (Z7)

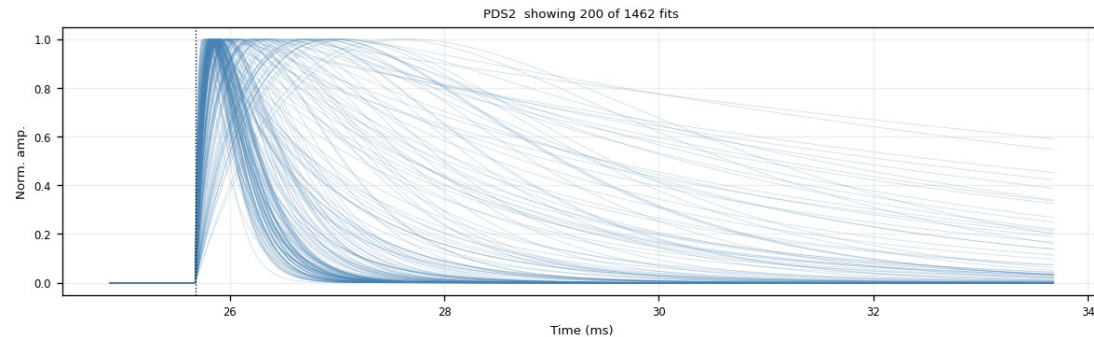
- Good channel PBS1: tight clean bundle — the $\tau_{\text{rise}} \leq 0.3$ ms cut removes $\sim 0\%$ (nearly free on quiet channels)
- Bad channel PDS2: broad, messy fan — the channel itself is bad; the cut trims the slow-rise curves, 1462 $\rightarrow$ 1154 (21%)



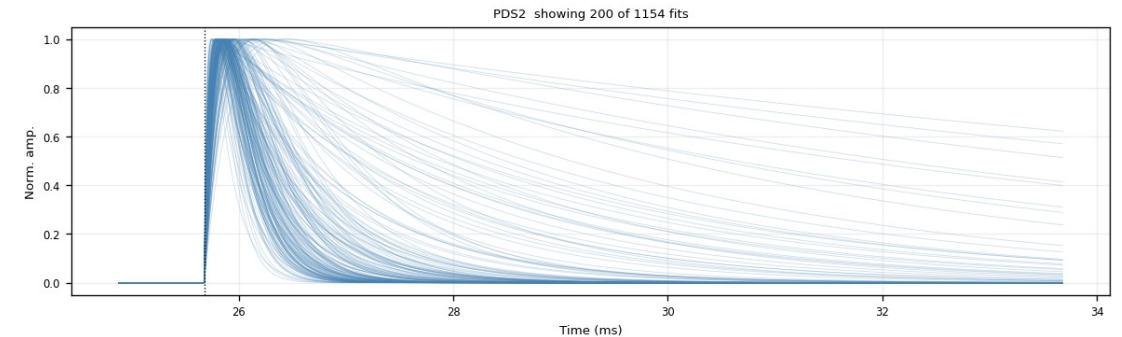
PBS1 (good) — $\text{NRMSE} \leq 0.4$



PBS1 (good) — $+\tau_{\text{rise}} \leq 0.3$ ms (unchanged)



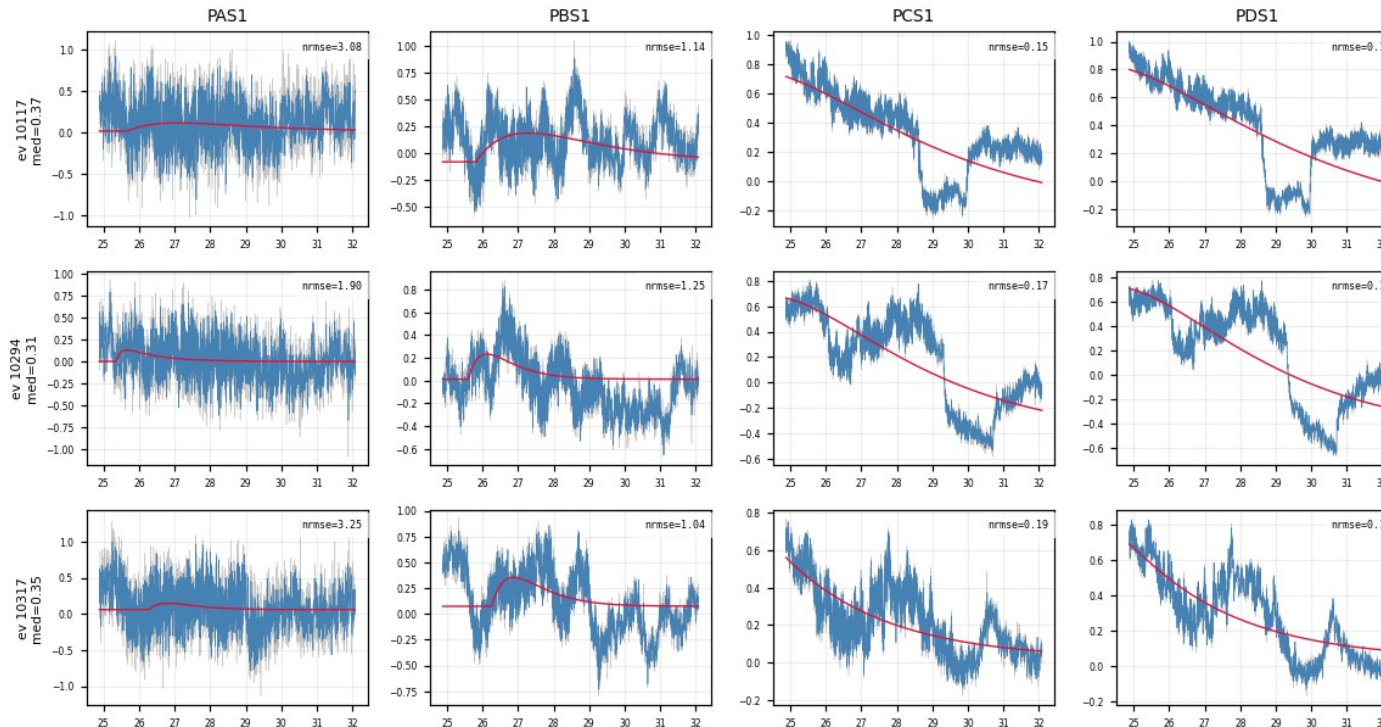
PDS2 (bad) — $\text{NRMSE} \leq 0.4$



PDS2 (bad) — $+\tau_{\text{rise}} \leq 0.3$ ms (slow risers trimmed)

Backup — what the τ_{rise} cut removes: noise NRMSE missed (Z22)

- Exactly the events that $\text{PASS NRMSE} \leq 0.4$ but are REMOVED by $\tau_{\text{rise}} \leq 0.3$ ms — the noise 0.4 let through, 0.3 catches
- PCS1 / PDS1 pass at $\text{NRMSE} \approx 0.15$ (a slow 2-exp hugs a slow baseline drift) — but the raw trace has no clean fast pulse
- Trade-off: the same cut also trims genuine slow-rise pulses — documented, decision pending

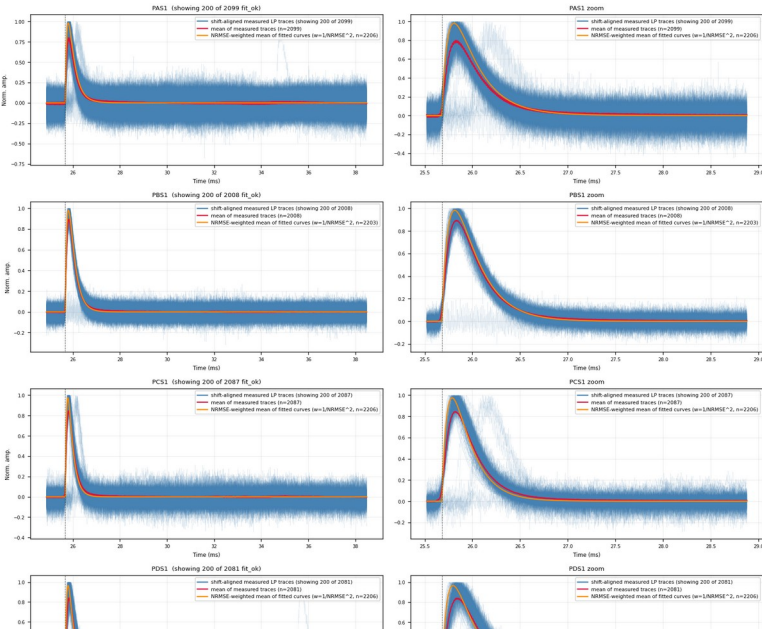


Z22: 3 events $\times$ PAS1/PBS1/PCS1/PDS1 — pass $\text{NRMSE} \leq 0.4$, removed by $\tau_{\text{rise}} \leq 0.3$ ms; PCS1/PDS1 are pure slow drift

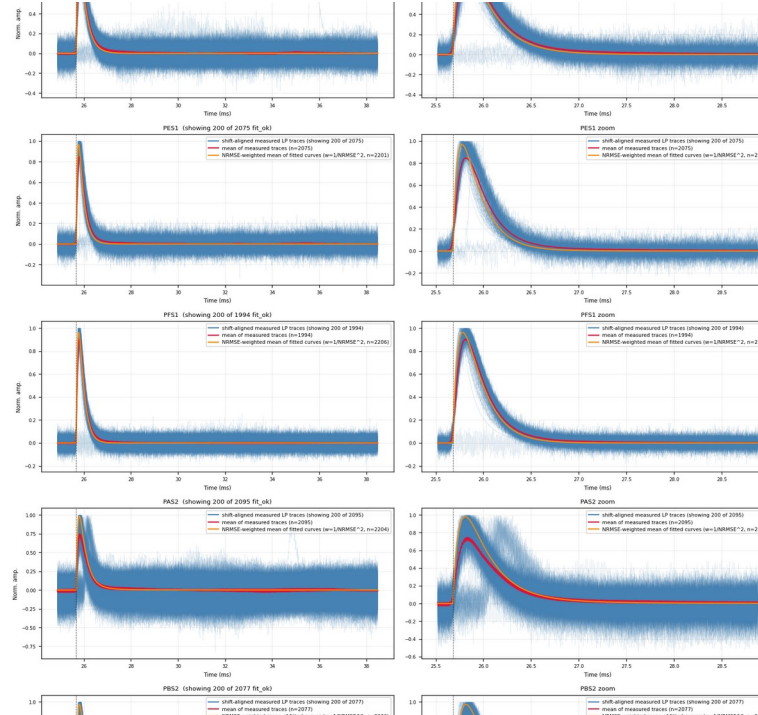
Backup gallery — Z7 · aligned_overlay — shift-aligned measured traces

Blue = measured LP traces shifted by (fitted pretrigger - 16050); red = point-by-point mean of ALL fit_ok events; orange = NRMSE-weighted mean of the fitted curves ($w = 1/\max(\text{NRMSE}, 0.01)^2$). Rise edges line up at 16050.

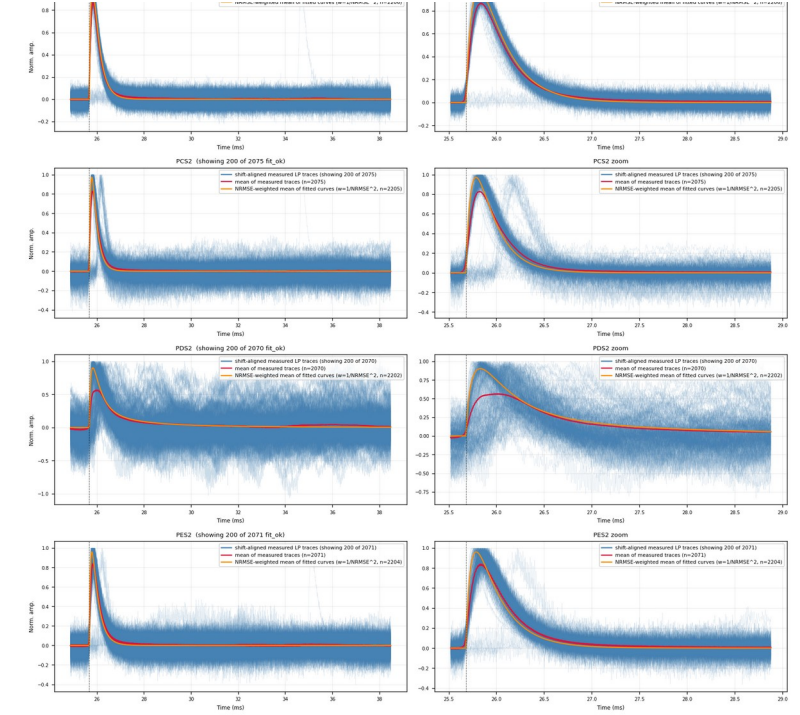
figure_10_fit_align_results.zip / 10_fit_aligned_overlay.png
Zip — 100kHz LP, shift-aligned measured traces (fit_ok only, no NRMSE cut)



part 1/3 (top to bottom)



part 2/3 (top to bottom)



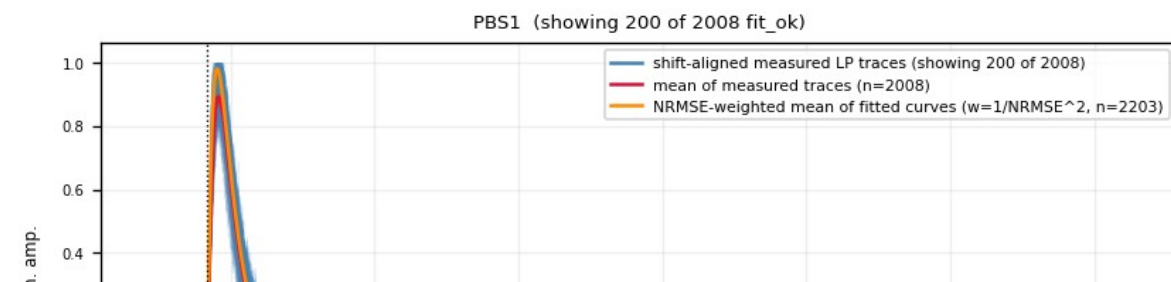
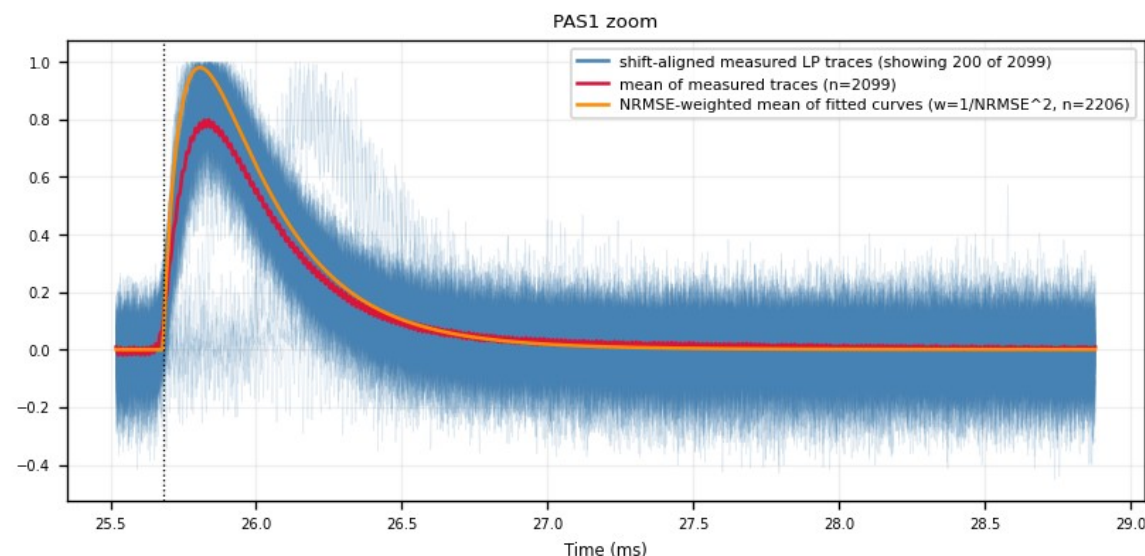
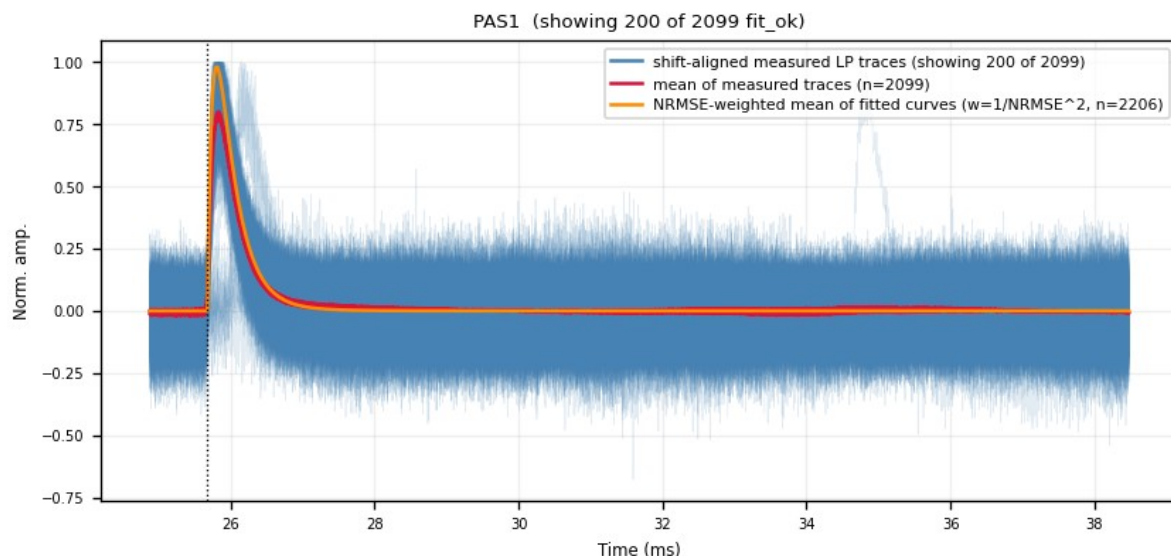
part 3/3 (top to bottom)

Backup gallery — Z7 · aligned_overlay — shift-aligned measured traces (detail, 1:1)

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
```

this figure: blue = shift-aligned MEASURED low-pass traces (display sample, up to 200/channel); red = point-by-point arithmetic mean of ALL N fit_ok shifted measured traces (fit contributes only the per-event shift, no fitted curve enters this mean); orange = NRMSE-WEIGHTED MEAN of the fitted 2-exp curves (each fit_ok event's fitted curve at common pretrigger 16050, peak-normalized, averaged with weight $w=1/\max(\text{NRMSE}, 0.01)^2$ so badly-fit events count less but are never excluded); dotted line = alignment reference 16050; fit_ok only, no NRMSE cut anywhere

Zip7 — 100kHz LP, shift-aligned measured traces (fit_ok only, no NRMSE cut)

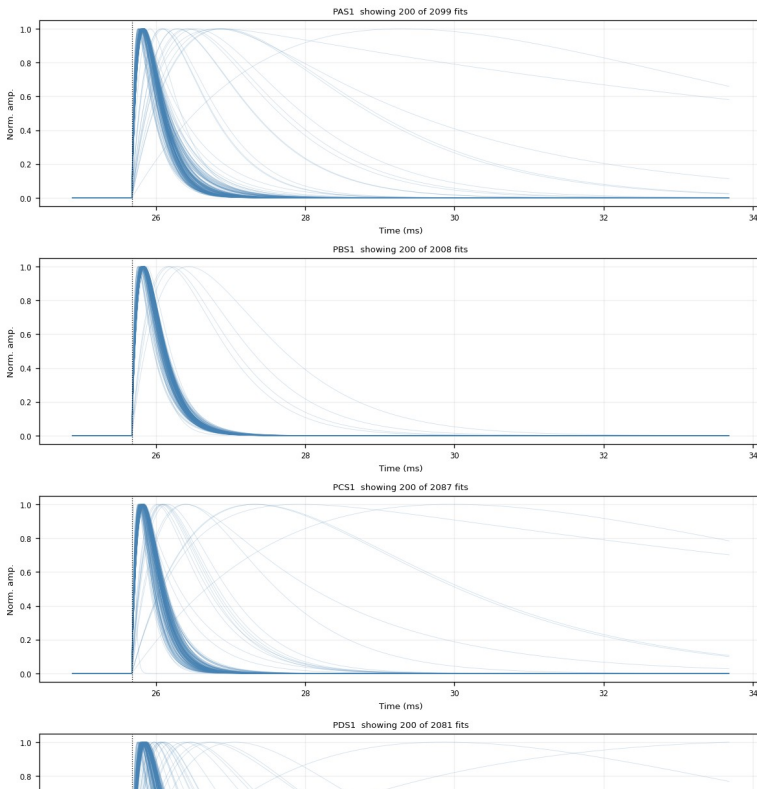


Backup gallery — Z7 · fitted_curves_overlay — fan of fitted 2-exp curves (no cut)

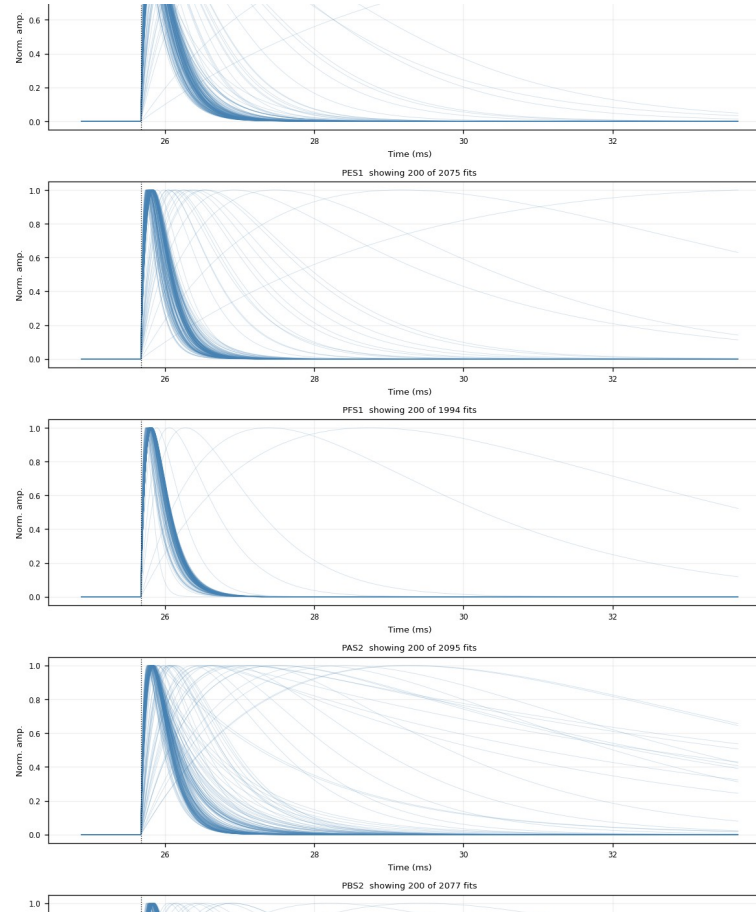
Every fit ok event's fitted curve re-evaluated at common pretrigger 16050, peak-normalized. Pure shape distribution; the broad slow curves are noise fits that the cuts remove.

```
processing: raw NIDA5 trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}}))$ ; ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+3000, maxrev=10000)
Fit ok => amp and dist (peak fall) -> NORMSE => RMS fit residual (fitted data peak, RECORDED ONLY (no cut))
-> align => shift the HEADRED LP trace by (fitted pretrigger - 16050); sub-sample up, interp
this figure: SMOOTH fitted 2-exp curves only (the required data); each fit ok event's fitted (amp,  $\tau_{\text{rise}}$ ,  $\tau_{\text{fall}}$ ) re-evaluated at
COMMON pretrigger=16050; peak-normalized; cut: fit ok, no NORMSE cut
```

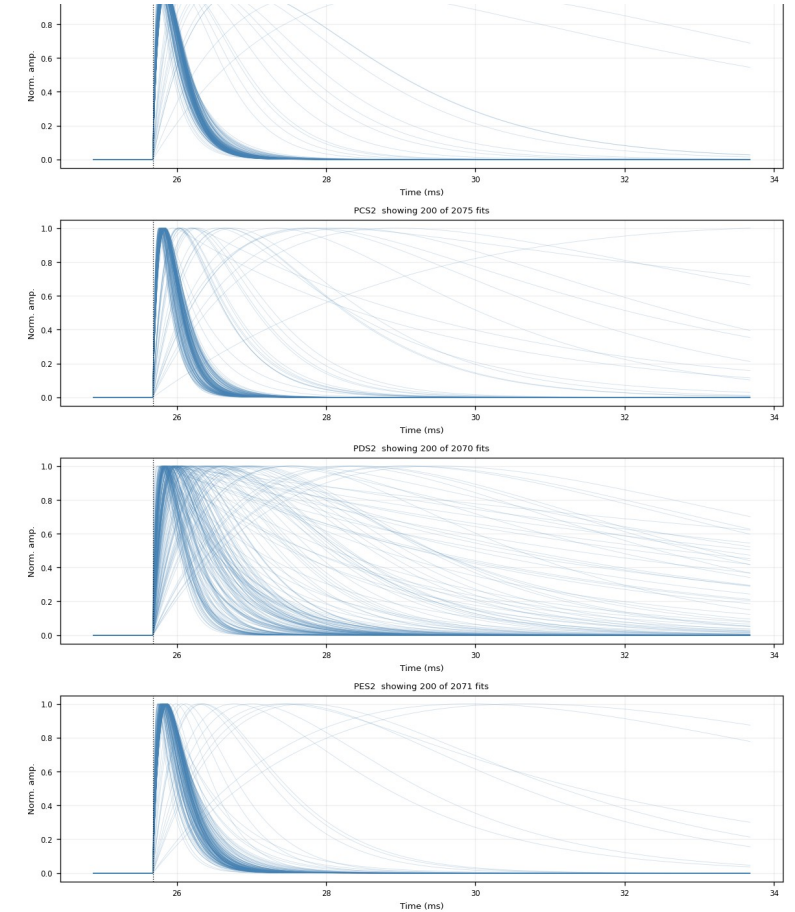
Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit ok, no NRMSE cut)



part 1/3 (top to bottom)



part 2/3 (top to bottom)



part 3/3 (top to bottom)

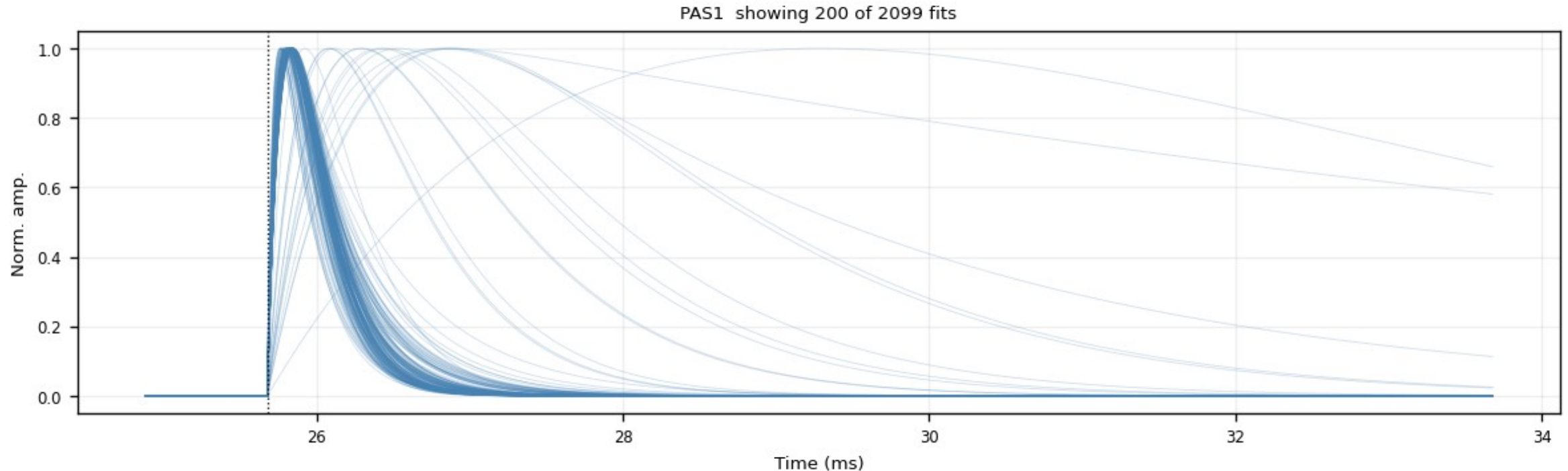
Backup gallery — Z7 · fitted_curves_overlay — fan of fitted 2-exp curves (no cut) (detail, 1:1)

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP_trace by (fitted pretrigger - 16050), sub-sample np.interp
```

this figure: SMOOTH fitted 2-exp curves only (no measured data): each fit_ok event's fitted (amp, t_rise, t_fall) re-evaluated at

COMMON pretrigger=16050, peak-normalized; cut: fit_ok, no NRMSE cut

Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, no NRMSE cut)



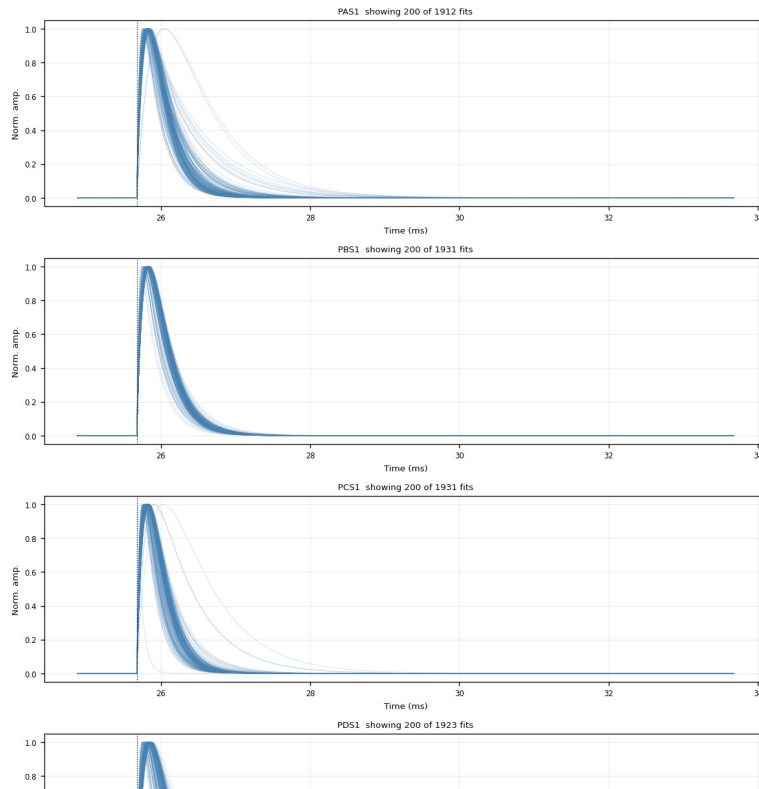
Backup gallery — Z7 · fitted_curves_overlay — after $\text{NRMSE} \leq 0.4$

figure: lp_fit_align results — zip7_fitted_curves_overlay_nrmse0.4.png

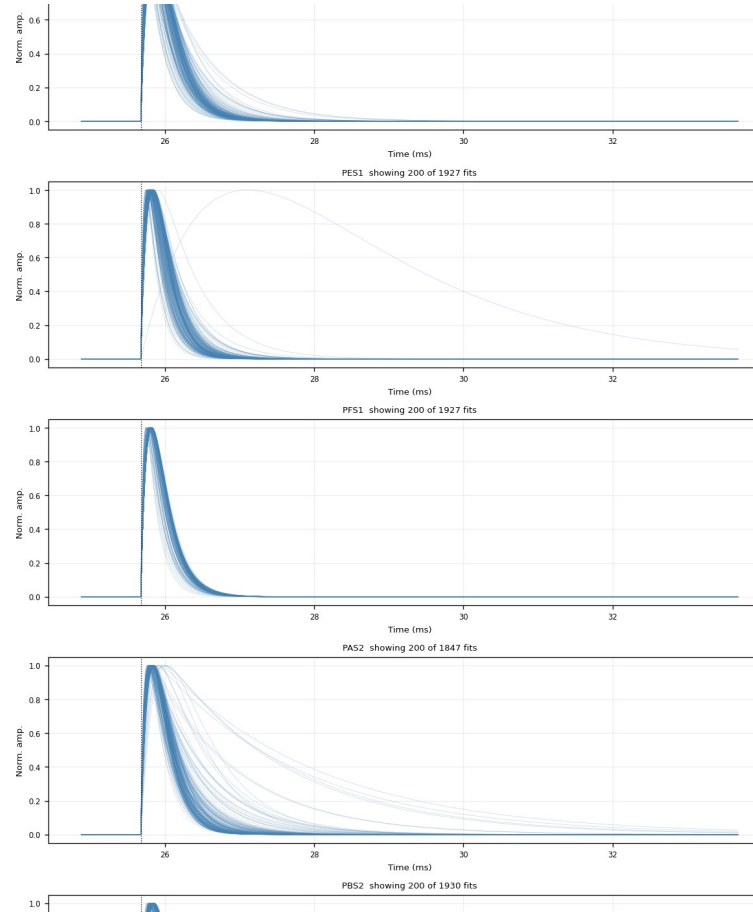
Same fan after the NRMSE cut: the noise fan collapses; the fast-pulse bundle remains.

```
processing: raw NIDA5 trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/\tau_{\text{fall}}) + \exp(-t/\tau_{\text{rise}}))$ ; ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxrev=10000)
Fit ok => amp and 0.5 decay fall -> NRMSE => RMS fit residual / fitted data peak. RECORDED ONLY (no cut)
-> align => shift the HEADRED LP trace by (fitted pretrigger - 16050); sub-sample up, interp
this figure: SMOOTH fitted 2-exp curves only (the required data); each fit ok event's fitted (amp,  $\tau_{\text{rise}}$ ,  $\tau_{\text{fall}}$ ) re-evaluated at
COMMON pretrigger=16050; peak-normalized; cut: fit ok, NRMSE<=0.4
```

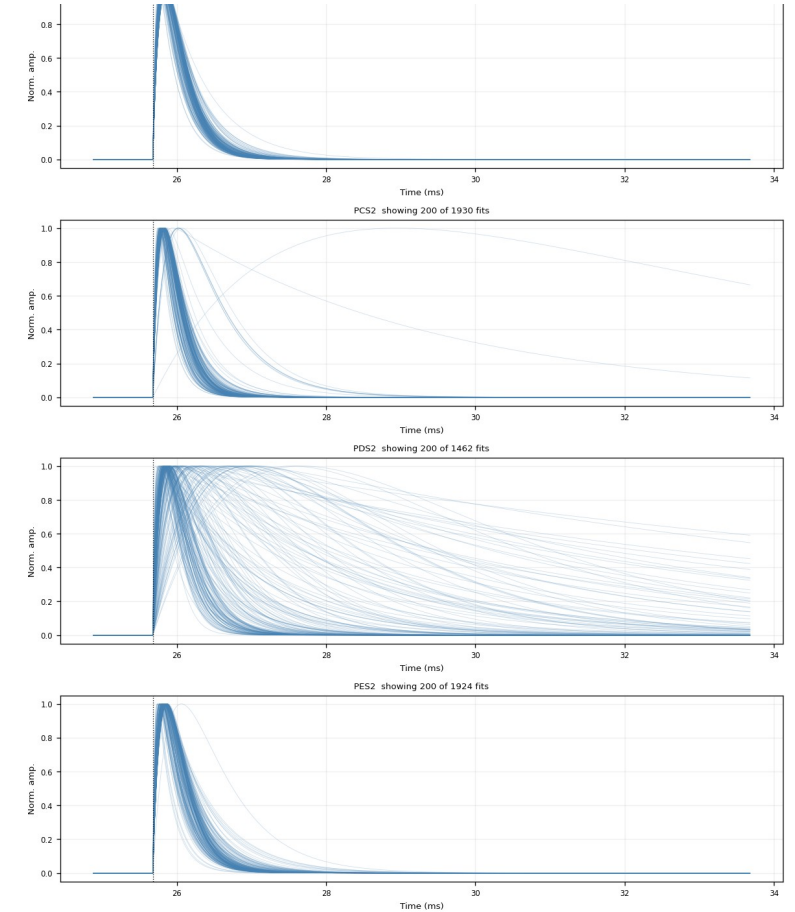
Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, NRMSE<=0.4)



part 1/3 (top to bottom)



part 2/3 (top to bottom)



part 3/3 (top to bottom)

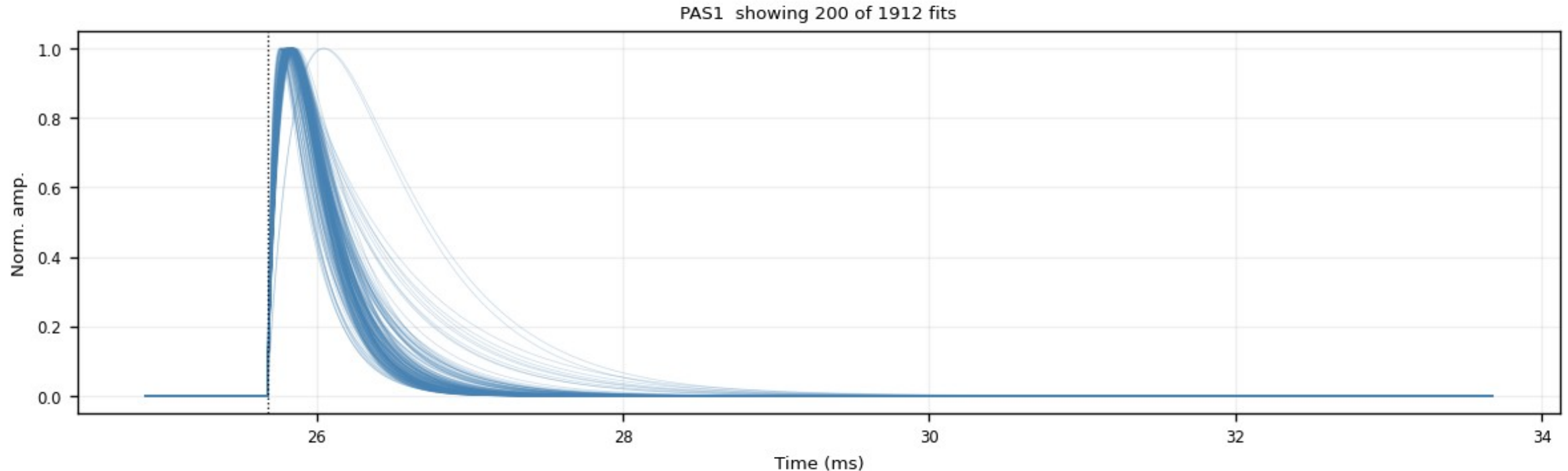
Backup gallery — Z7 · fitted_curves_overlay — after $\text{NRMSE} \leq 0.4$ (detail, 1:1)

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP_trace by (fitted pretrigger - 16050), sub-sample np.interp
```

this figure: SMOOTH fitted 2-exp curves only (no measured data): each fit_ok event's fitted (amp, t_rise, t_fall) re-evaluated at

COMMON pretrigger=16050, peak-normalized; cut: fit_ok, $\text{NRMSE} \leq 0.4$

Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, $\text{NRMSE} \leq 0.4$)



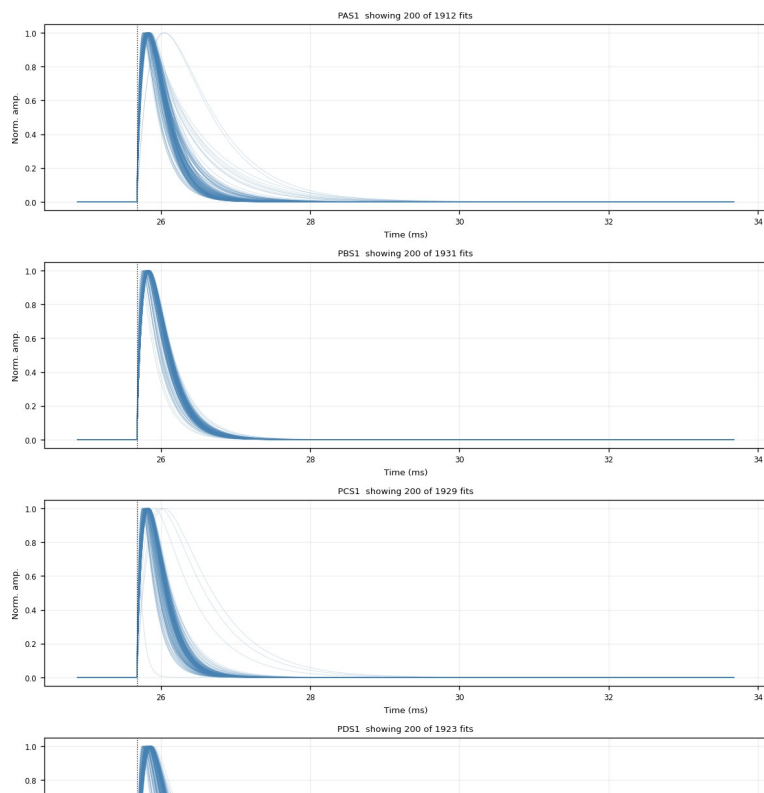
Backup gallery — Z7 · fitted_curves_overlay — after $\text{NRMSE} \leq 0.4$ and $\tau_{\text{rise}} \leq 0.3 \text{ ms}$

The exact PCA input population: clean fast-pulse bundle only.

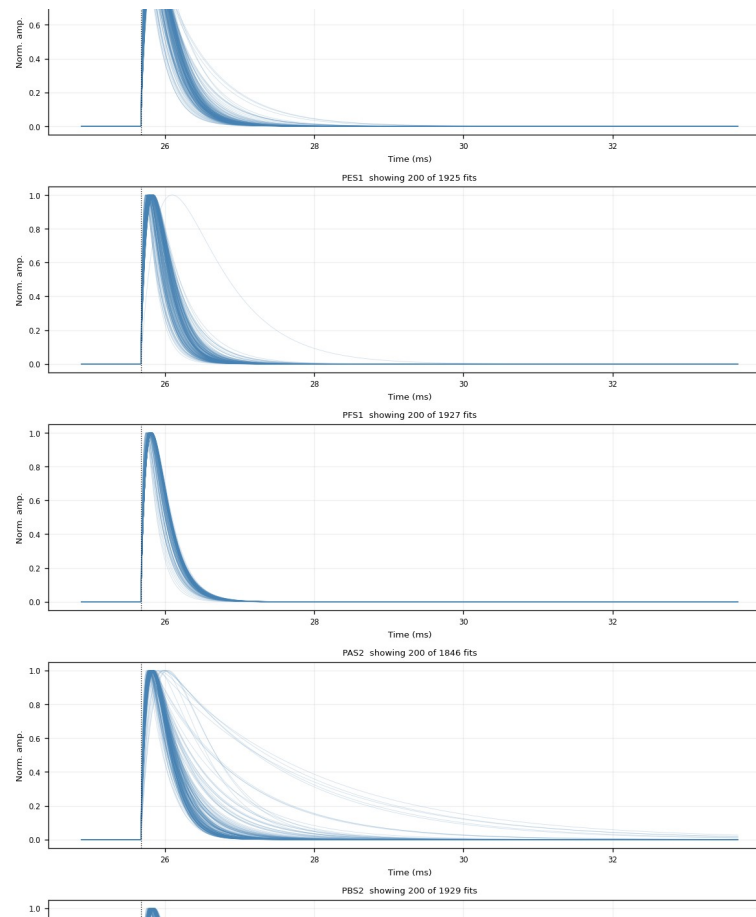
figure: ip_fit_align results — zip7_fitted_curves_overlay_nrmse0.4_trise0.30ms.png

```
processing: raw NIDA5 trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}}))$ ; ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxrev=10000)
Fit ok => amp0 and t_rise0 fall > NRMSE => NRMSE fit residuals/fitted data peak, RECORDED ONLY (no cut)
-> align => shift the HEADRED LP trace by (fitted pretrigger - 16050); sub-sample np.integ
this figure: SMOOTH fitted 2-exp curves only (the measured data); each fit is event's fitted (amp, t_rise, t_fall) re-evaluated at
COMMON pretrigger=16050, peak-normalized; cut: fit_ok, NRMSE<=0.4, t_rise<=0.30ms
```

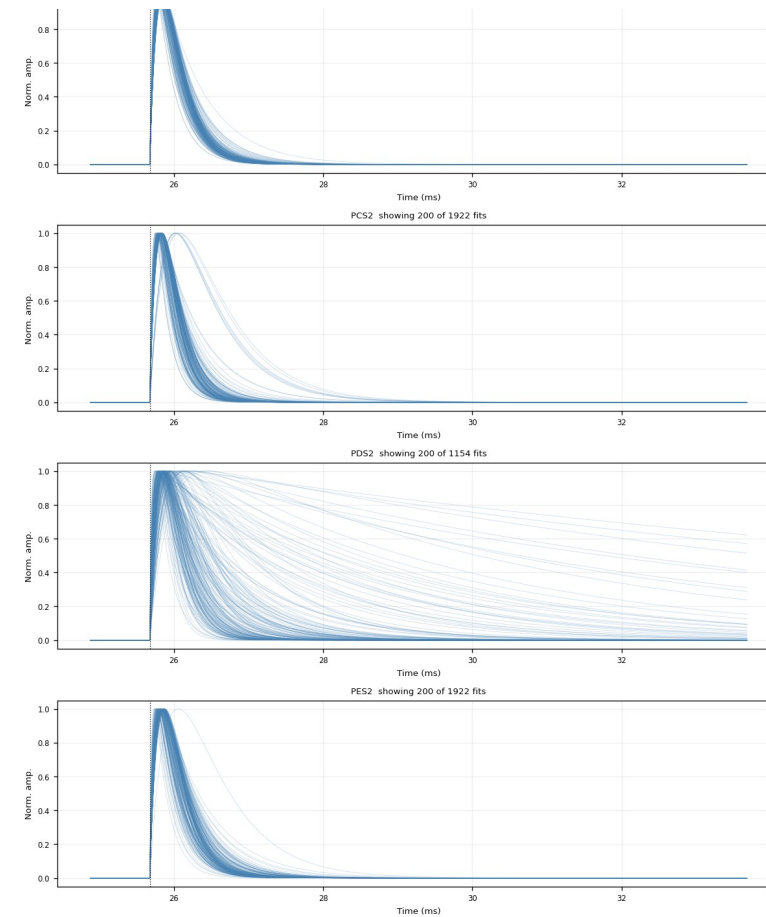
Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, NRMSE<=0.4, $\tau_{\text{rise}} \leq 0.30 \text{ ms}$)



part 1/3 (top to bottom)



part 2/3 (top to bottom)

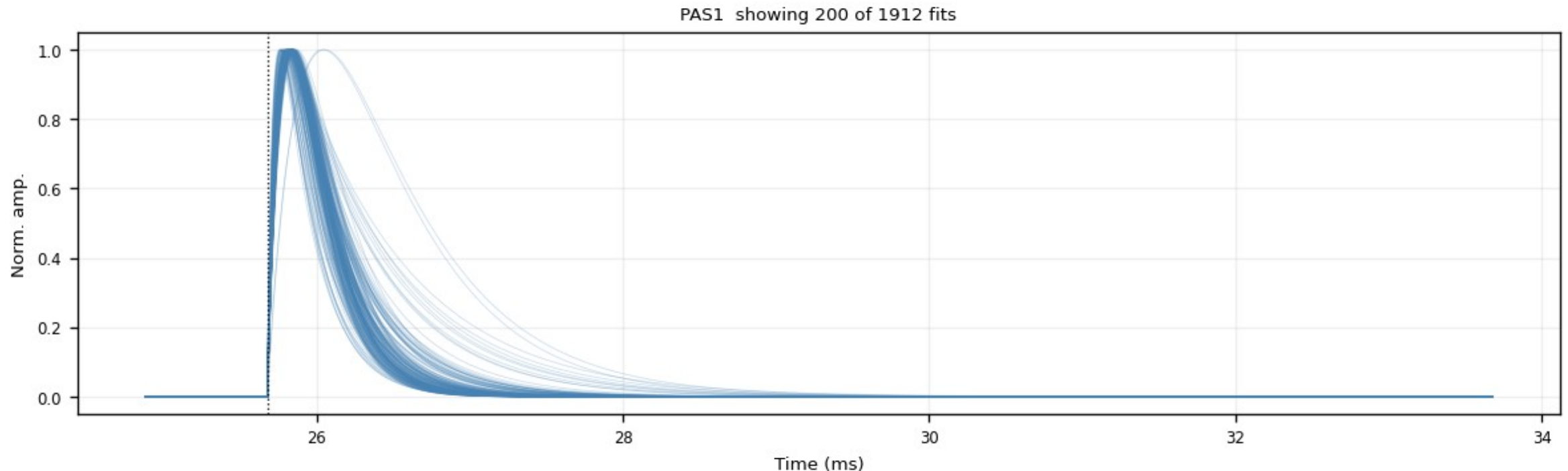


part 3/3 (top to bottom)

Backup gallery — Z7 · fitted_curves_overlay — after $\text{NRMSE} \leq 0.4$ and $\tau_{\text{rise}} \leq 0.3 \text{ ms}$ (detail, 1:1)

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and  $0 < t_{\text{rise}} < t_{\text{fall}}$  ->  $\text{NRMSE} := \text{RMS}(\text{fit residual}) / \text{fitted pulse peak}$ , RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: SMOOTH fitted 2-exp curves only (no measured data): each fit_ok event's fitted (amp,  $t_{\text{rise}}$ ,  $t_{\text{fall}}$ ) re-evaluated at
COMMON pretrigger=16050, peak-normalized; cut: fit_ok,  $\text{NRMSE} \leq 0.4$ ,  $t_{\text{rise}} \leq 0.30 \text{ ms}$ 
```

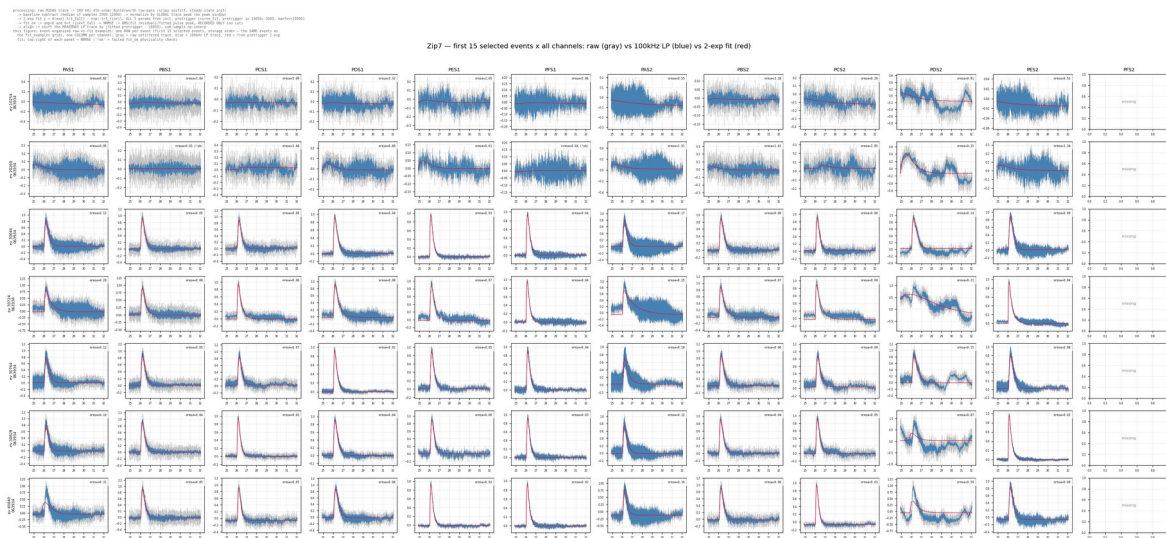
Zip7 — fitted 2-exp curves, common pretrigger=16050, peak-normalized (fit_ok, $\text{NRMSE} \leq 0.4$, $t_{\text{rise}} \leq 0.30 \text{ ms}$)



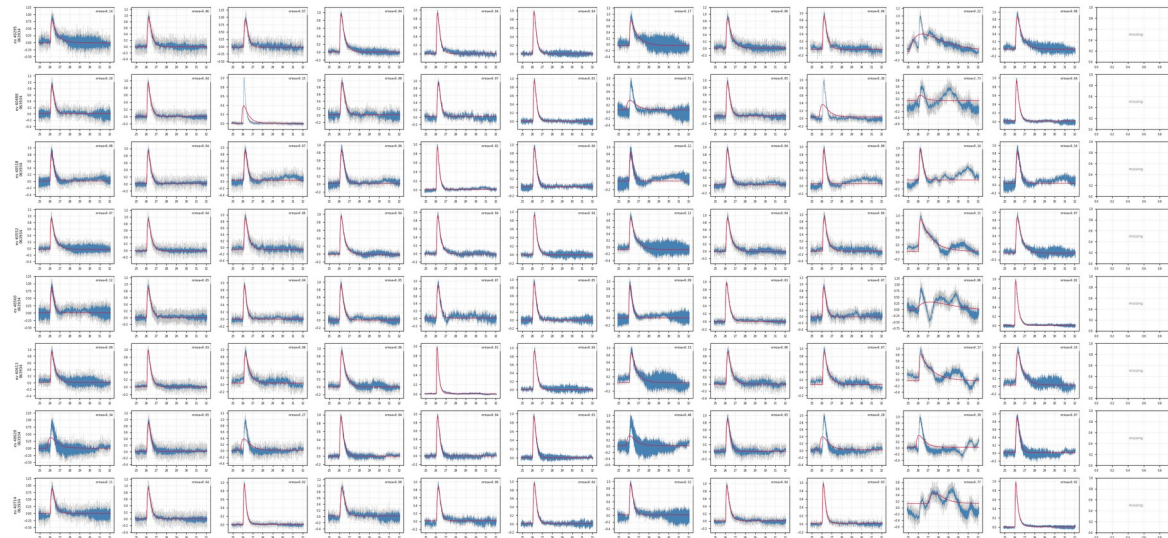
Backup gallery — Z7 · raw_vs_fit_examples — first 15 events × all channels

Gray = raw unfiltered, blue = 100 kHz LP, red = 2-exp fit; per-panel NRMSE top-right. Same 15 events as fit_examples; noise vs pulse size is directly visible.

figure: ip_fit_align results — zip7_raw_vs_fit_examples.png



part 1/2 (top to bottom)

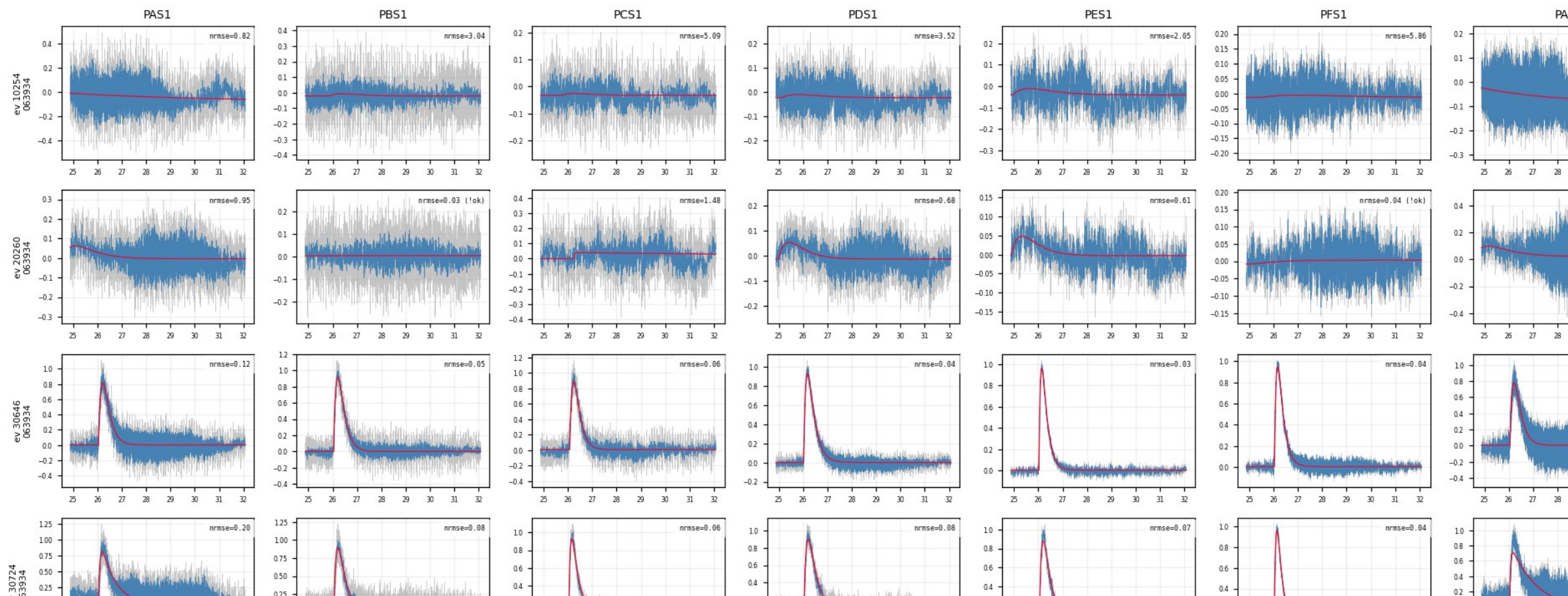


part 2/2 (top to bottom)

Backup gallery — Z7 · raw_vs_fit_examples — first 15 events × all channels (detail, 1:1)

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
 -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
 -> 2-exp fit $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$, ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxfev=10000)
 -> fit_ok := amp>0 and 0<t_rise< fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
 -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
 this figure: event-organized raw-vs-fit examples: one ROW per event (first 15 selected events, storage order - the SAME events as the fit_examples grid), one COLUMN per channel; gray = raw unfiltered trace, blue = 100kHz LP trace, red = free-pretrigger 2-exp fit; top-right of each panel = NRMSE ('!ok' = failed fit_ok physicality check)

Zip7 — first 15 selected events x all channels: raw (gray) vs 100

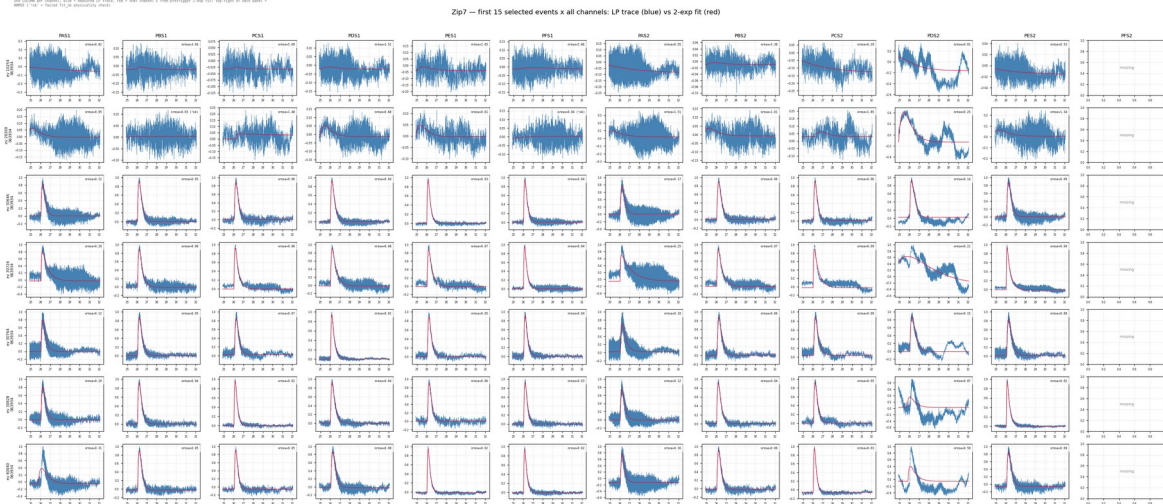


Backup gallery — Z7 · fit_examples — first 15 events × all channels

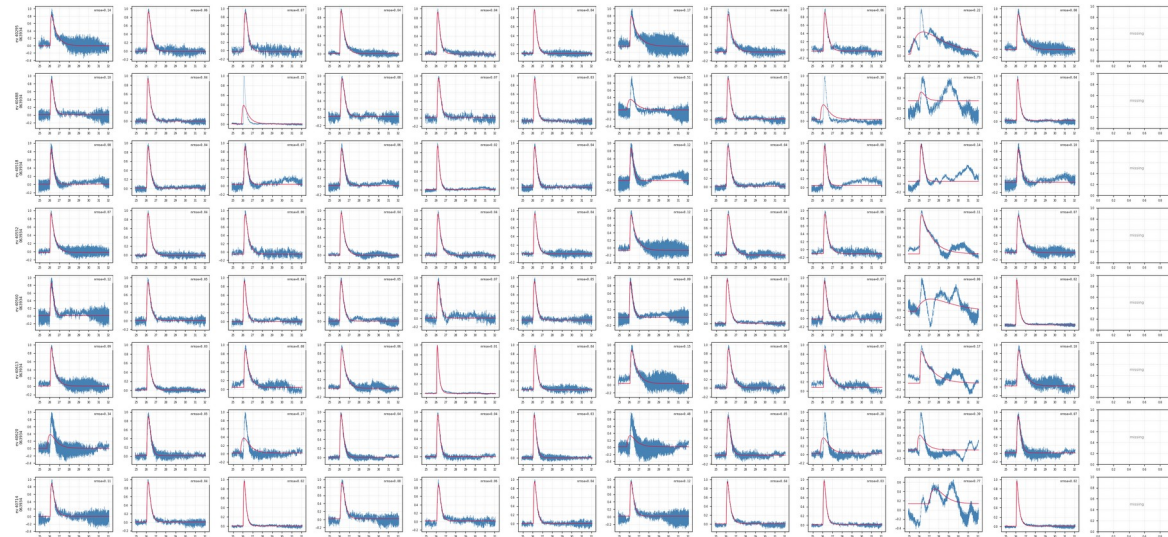
(LP vs fit)

One ROW per event, one COLUMN per channel. A real event shows a clean pulse and a good fit in every channel simultaneously; a noise trigger fails across the whole row.

figure: lp_fit_align_results — zip7_fit_examples.png



part 1/2 (top to bottom)

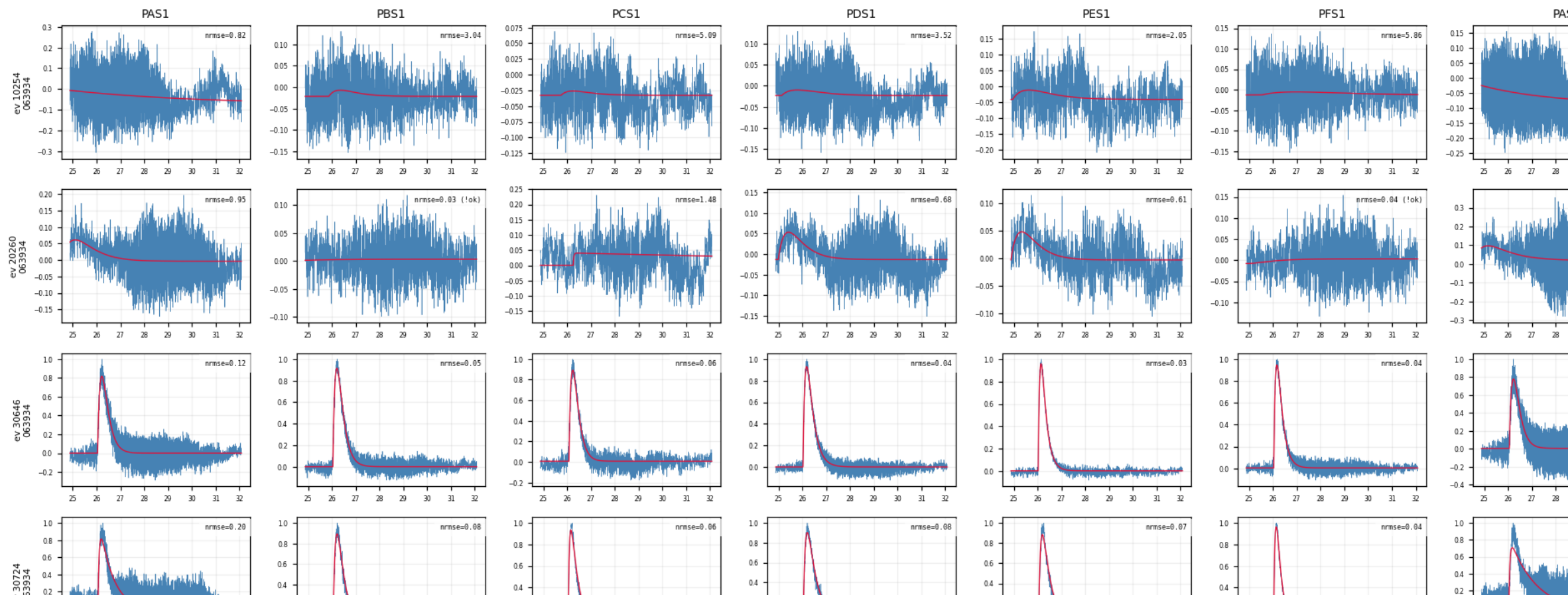


part 2/2 (top to bottom)

Backup gallery — Z7 · fit_examples — first 15 events × all channels (LP vs fit) (detail, 1:1)

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}}))$, ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxfev=10000)
-> fit ok := amp>0 and 0<=t_rise<=t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: event-organized fit examples: one ROW per event (first 15 selected events in storage order, NOT quality-selected),
one COLUMN per channel; blue = measured LP trace, red = that channel's free-pretrigger 2-exp fit; top-right of each panel =
NRMSE ('!ok' = failed fit_ok physicality check)

Zip7 — first 15 selected events x all channels: LP trace (

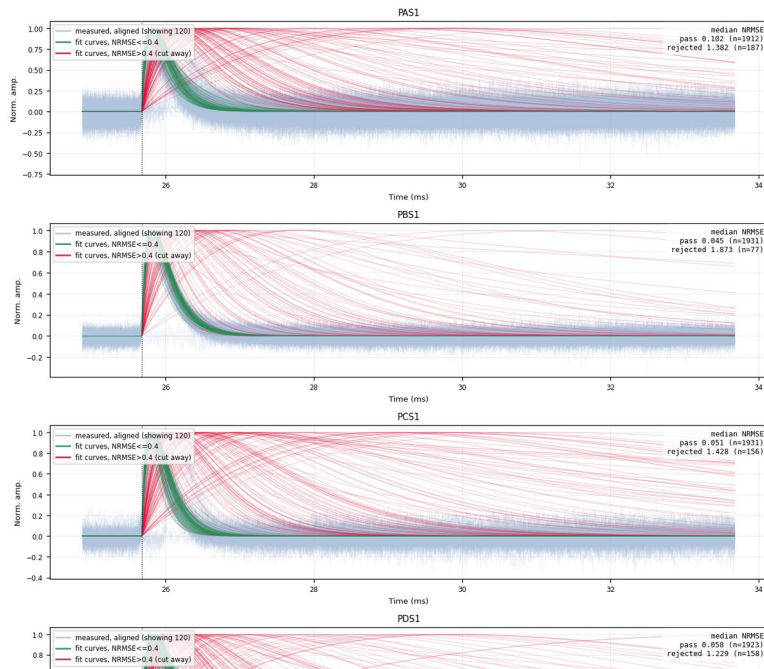


Backup gallery — Z7 · overlay_fan_cut — measured traces + kept/rejected fits

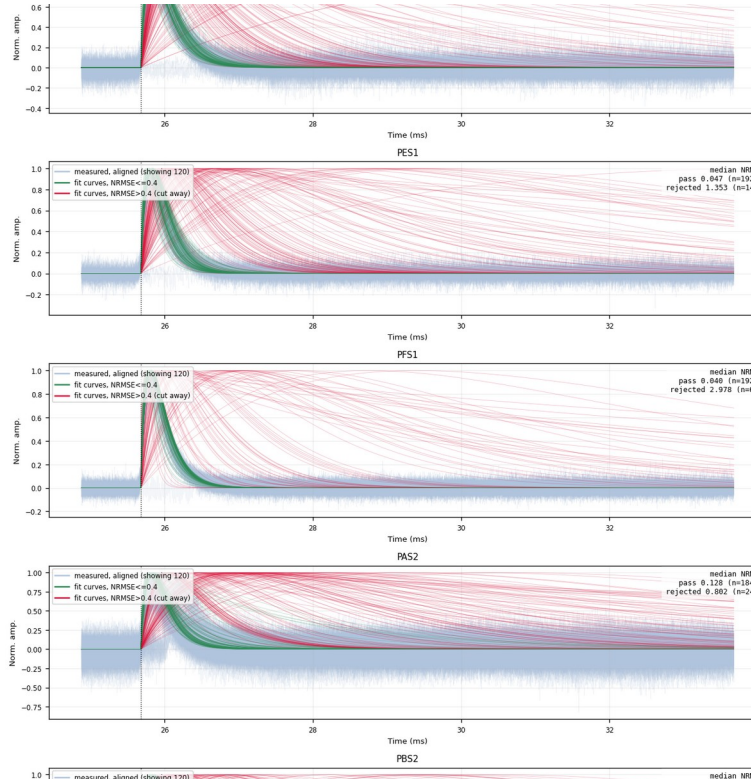
Gray-blue = aligned measured traces; green = fitted curves with $\text{NRMSE} \leq 0.4$ (kept); red = $\text{NRMSE} > 0.4$ (cut away). Median NRMSE and counts of both populations figure. 1p fit_align_results — zip7_overlay_fan_cut_nrmse0.4.png stamped per channel.

```
processing: raw NIDA6 trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GABA trace peak (no peak visible)
-> 2-exp fit y = A*(exp(-(t-T)/tau1)) + exp(-(t-T)/tau2), ALL 5 param free incl. pretrigger (curve fit, pretrigger in 16000-3000, maxiter=10000)
-> fit ok => amp and det (line fit) => NRMSE = RMSE(fit residual)/fitted value peak, RECORDED ONLY (no cut)
-> align => shift the READOUT LP trace by (fitted pretrigger - 16000), sub-sample no interp
This figure: three views in one plot: short-aligned MEASURED LP traces (gray-blue, fit ok, 80 mean line) + SMOOTH fitted 2-exp curve at common pretrigger 16000, peak-normalized, salt by the NRMSE cut: green = NRMSE<=0.4 (kept), red = NRMSE>0.4 (cut away); top right box = median NRMSE and counts of each population
```

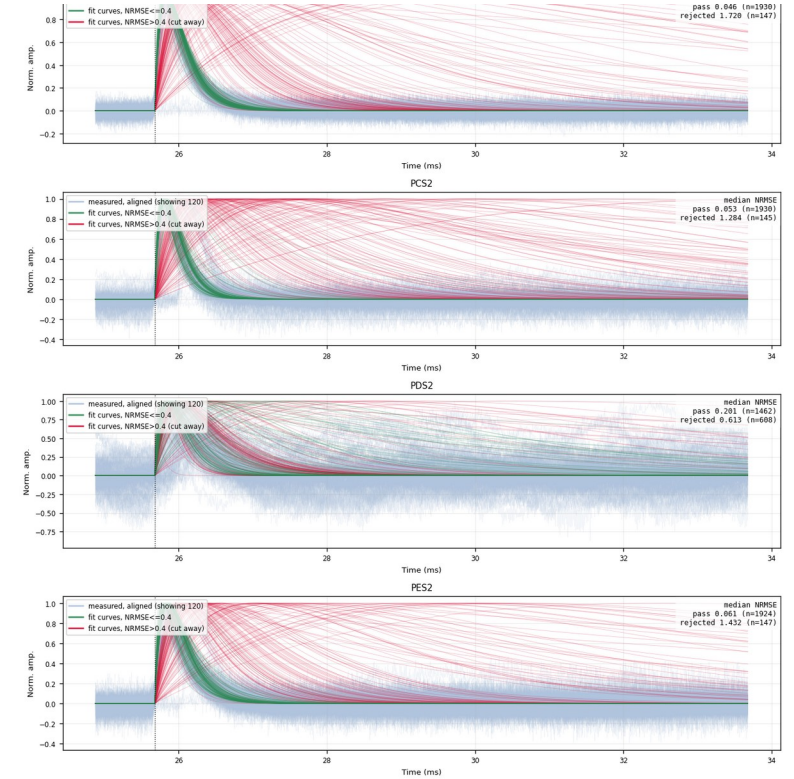
Zip7 — aligned measured traces + fitted curves split by NRMSE cut 0.4



part 1/3 (top to bottom)



part 2/3 (top to bottom)



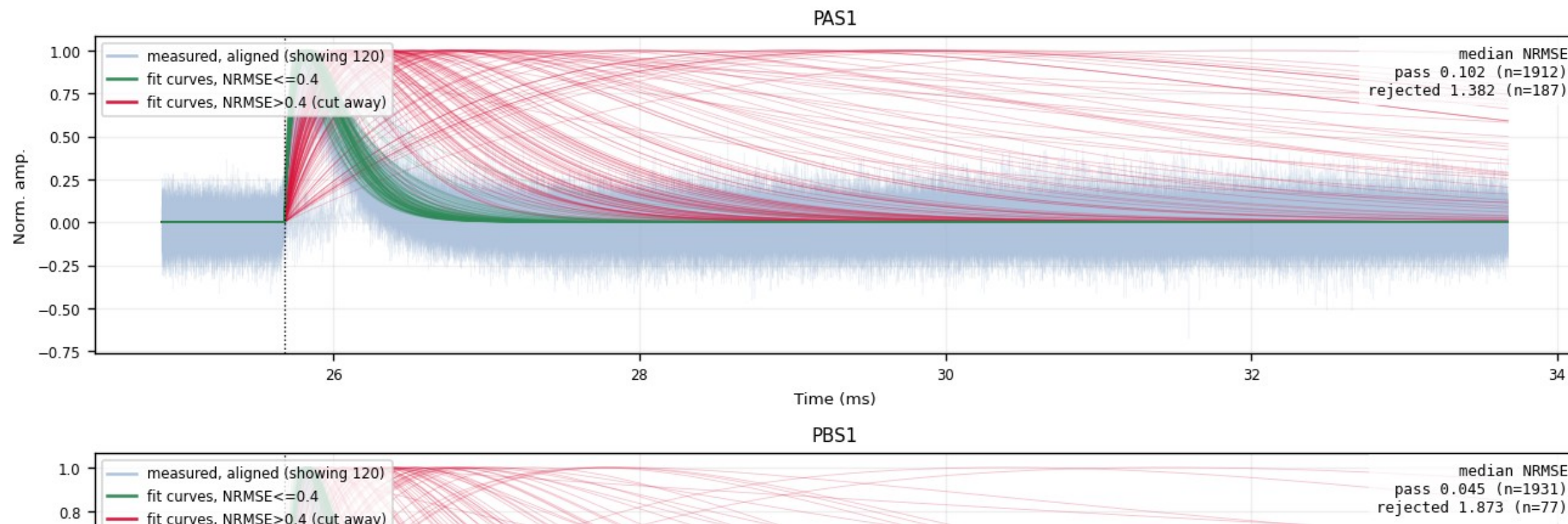
part 3/3 (top to bottom)

Backup gallery — Z7 · overlay_fan_cut — measured traces + kept/rejected fits (detail, 1:1)

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
```

this figure: three views in one panel: shift-aligned MEASURED LP traces (gray-blue, fit ok, NO mean line) + SMOOTH fitted 2-exp curves at common pretrigger 16050, peak-normalized, split by the NRMSE cut: green = NRMSE≤0.4 (kept), red = NRMSE>0.4 (cut away); top-right box = median NRMSE and counts of each population

Zip7 — aligned measured traces + fitted curves split by NRMSE cut 0.4



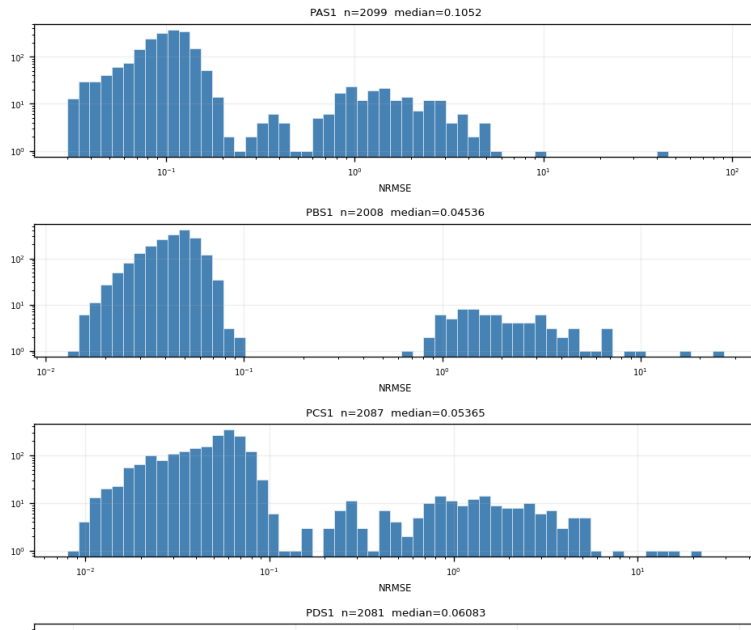
Backup gallery — Z7 · nrmse — NRMSE distribution (log-log)

figure: lp_fit_align results — zip7_nrmse.png

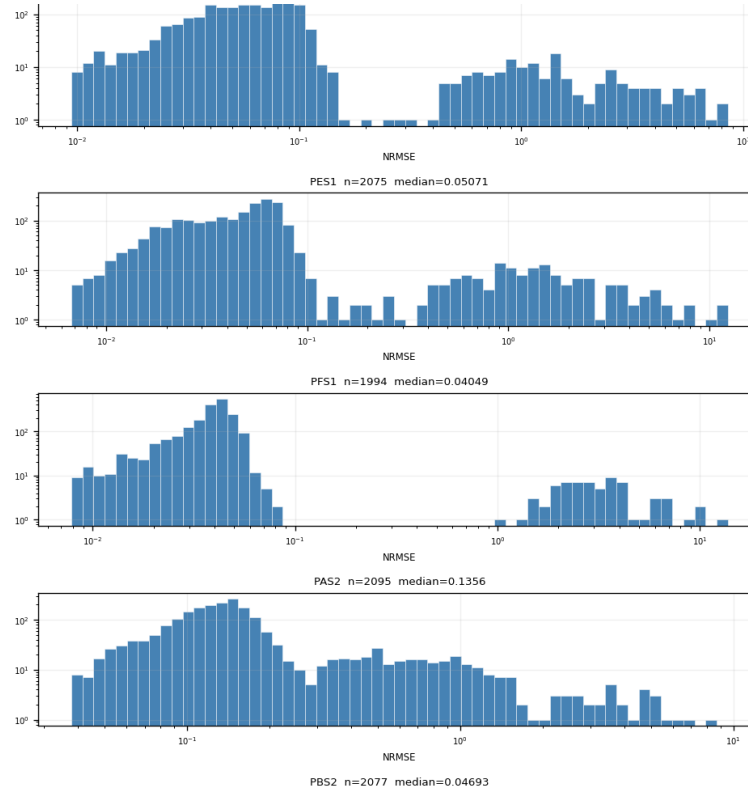
NRMSE = RMS(fit residual)/fitted peak. Bimodal: good-fit population ($\sim 0.05\text{--}0.1$) vs noise triggers ($\sim 1\text{--}2$); the valley at ≈ 0.4 sets the cut.

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+3000, maxfev=10000)
-> fit ok := amp>0 and Get Rise&Fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: NRMSE distribution of fit ok events, log-log axes (log-spaced bins, log counts); bimodal = good-fit population vs noise triggers; valley = natural cut candidate; NO cut applied
```

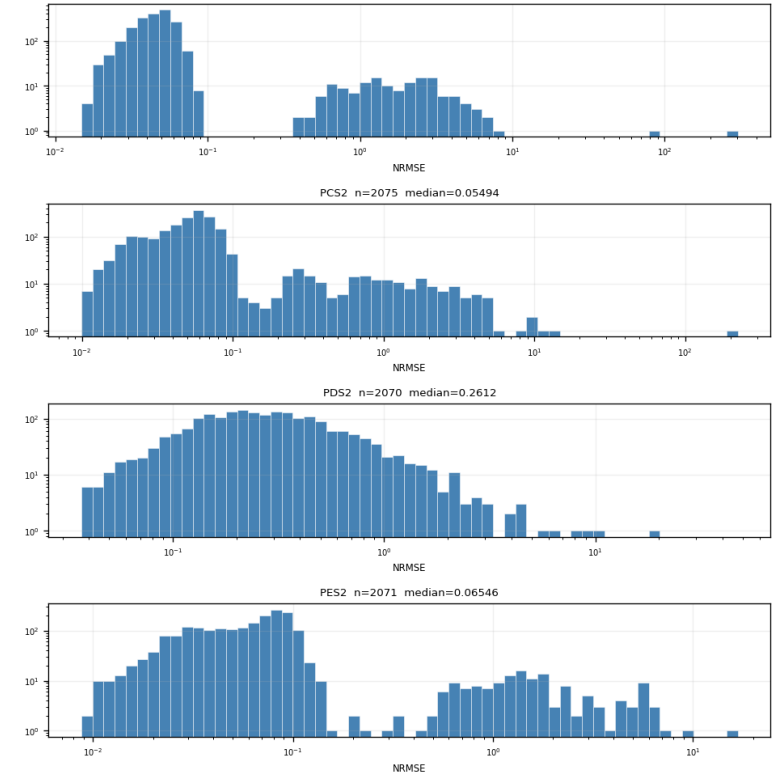
Zip7 — nrmse distribution (fit_ok events)



part 1/3 (top to bottom)



part 2/3 (top to bottom)

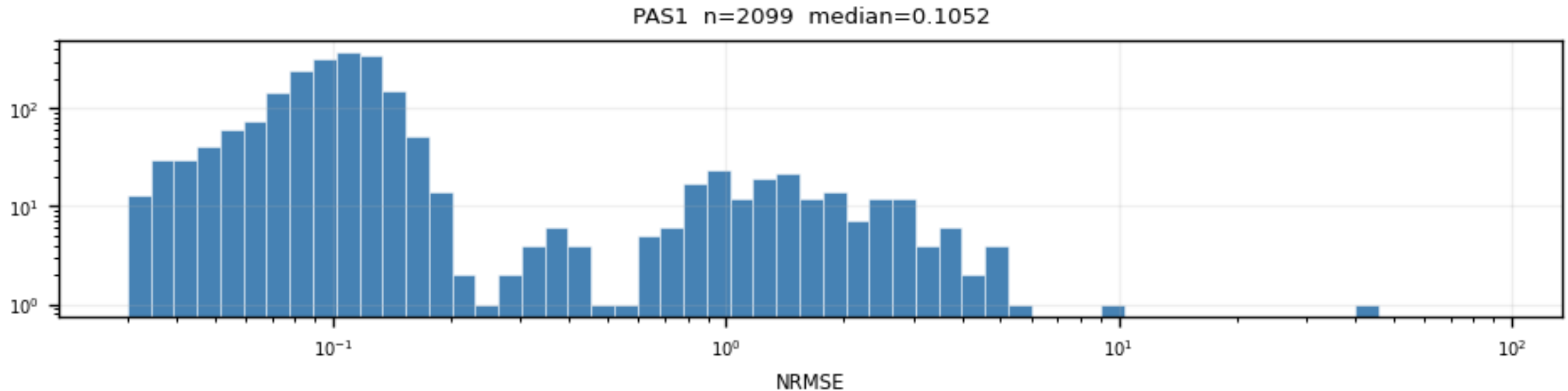


part 3/3 (top to bottom)

Backup gallery — Z7 · nrmse — NRMSE distribution (log-log) (detail, 1:1)

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit ok := amp>0 and  $0 < t_{\text{rise}} < t_{\text{fall}}$  -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: NRMSE distribution of fit_ok events, log-log axes (log-spaced bins, log counts); bimodal = good-fit population vs
noise triggers; valley = natural cut candidate; NO cut applied
```

Zip7 — nrmse distribution (fit_ok events)



PRS1 n=2008 median=0.04536

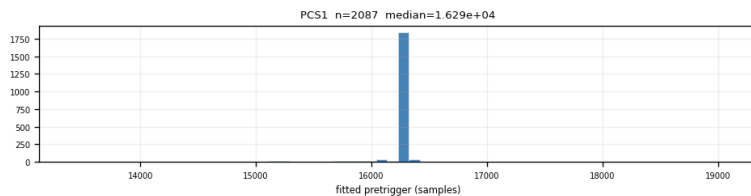
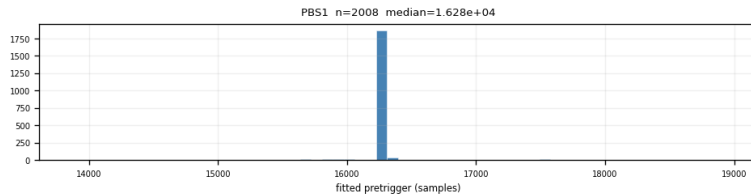
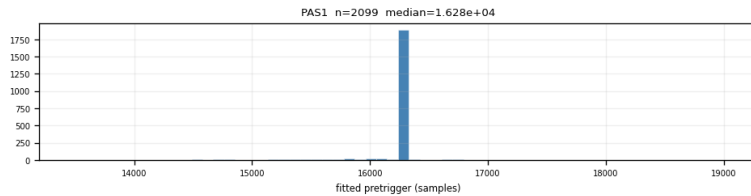
Backup gallery — Z7 · pretrigger — fitted onset distribution

figure: lp_fit_align results — zip7_pretrigger.png

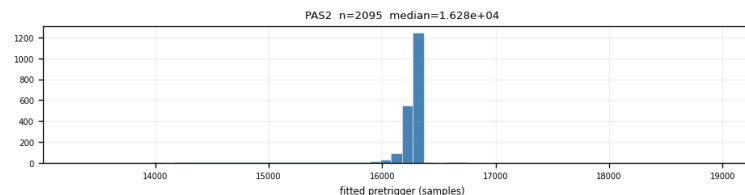
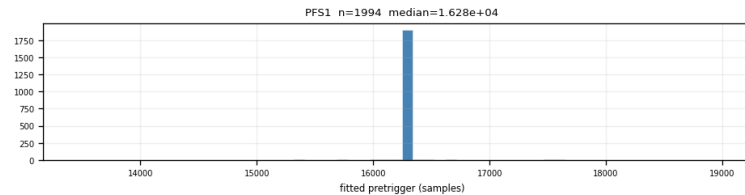
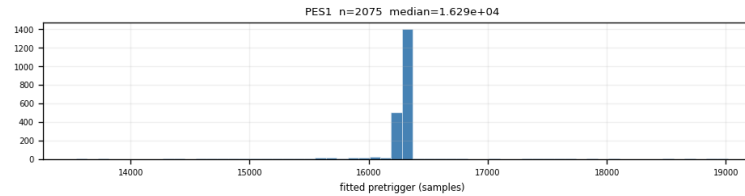
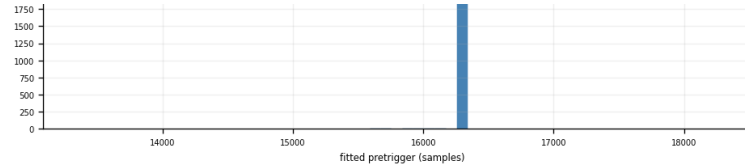
The onset is a FREE fit parameter; fitted values cluster ≈ 230 samples after the nominal 16050, which is why pinning it inside the fit was wrong.

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A \exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}})$ . ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxfev=10000)
-> fit ok := amp>0 and 0<=rise<fall -> RMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: distribution of the FITTED free pretrigger (samples) for fit_ok events;
```

Zip7 — pretrigger distribution (fit_ok events)

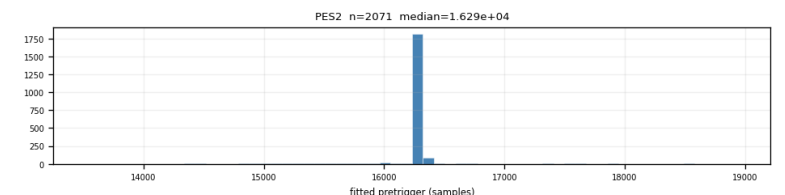
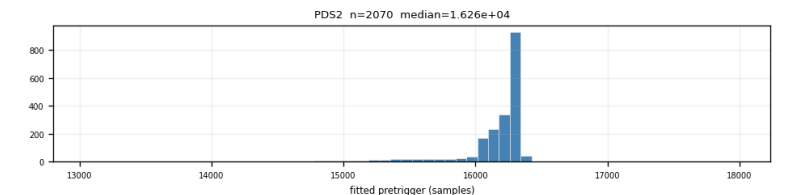
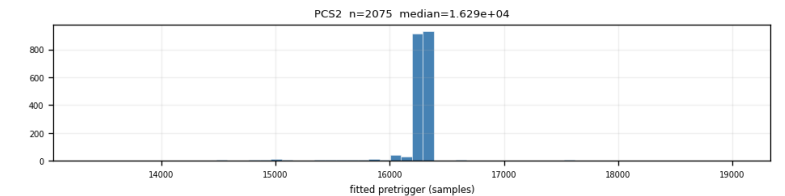
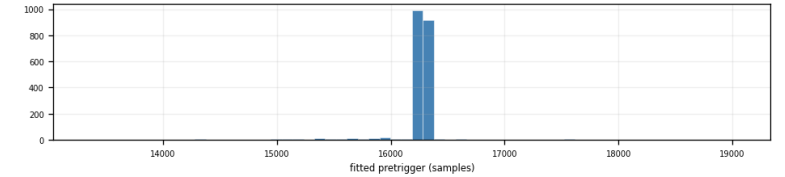


part 1/3 (top to bottom)



PBS2 n=2077 median=1.628e+04

part 2/3 (top to bottom)

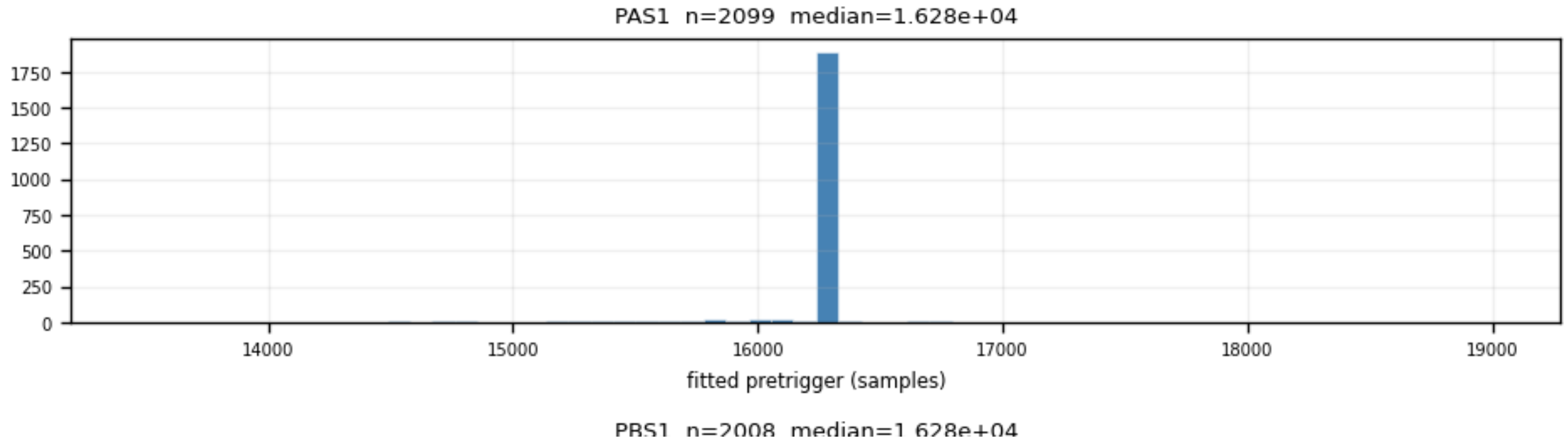


part 3/3 (top to bottom)

Backup gallery — Z7 · pretrigger — fitted onset distribution (detail, 1:1)

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit ok := amp>0 and  $0 < t_{\text{rise}} < t_{\text{fall}}$  -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: distribution of the FITTED free pretrigger (samples) for fit_ok events; reference 16050
```

Zip7 — pretrigger distribution (fit_ok events)



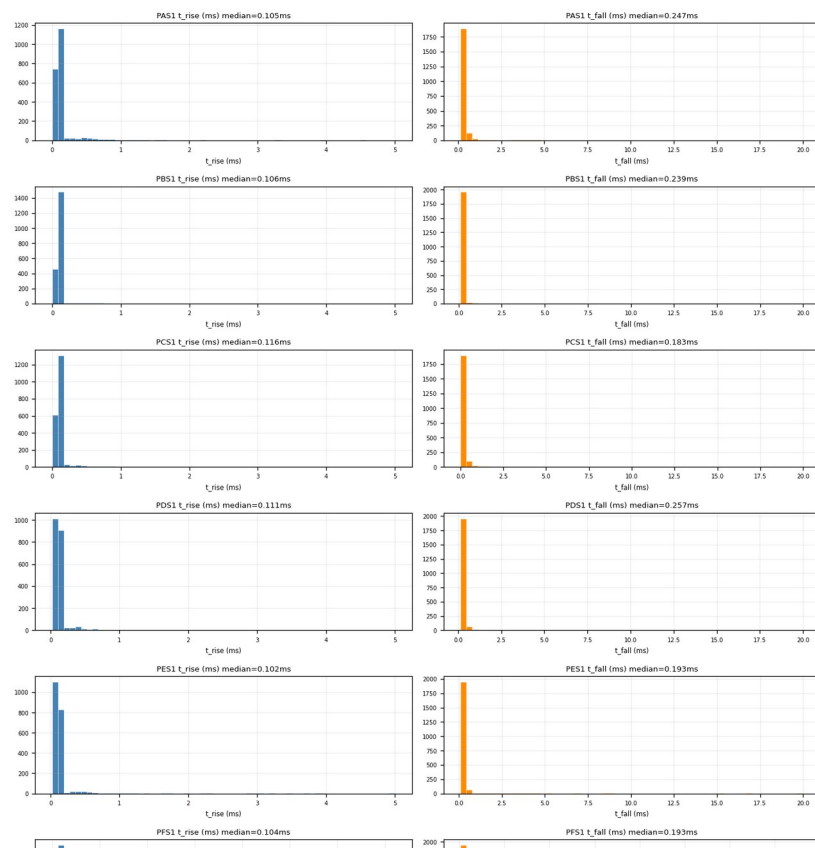
Backup gallery — Z7 · time_constants — fitted τ_{rise} / τ_{fall} distributions

Fast-pulse population sits at $\tau_{\text{rise}} \approx 0.1$ ms; isolated spikes at the parameter bounds (5 ms / 20 ms) are fits pinned at the fit limits (noise), removed by the NRMSE cut.

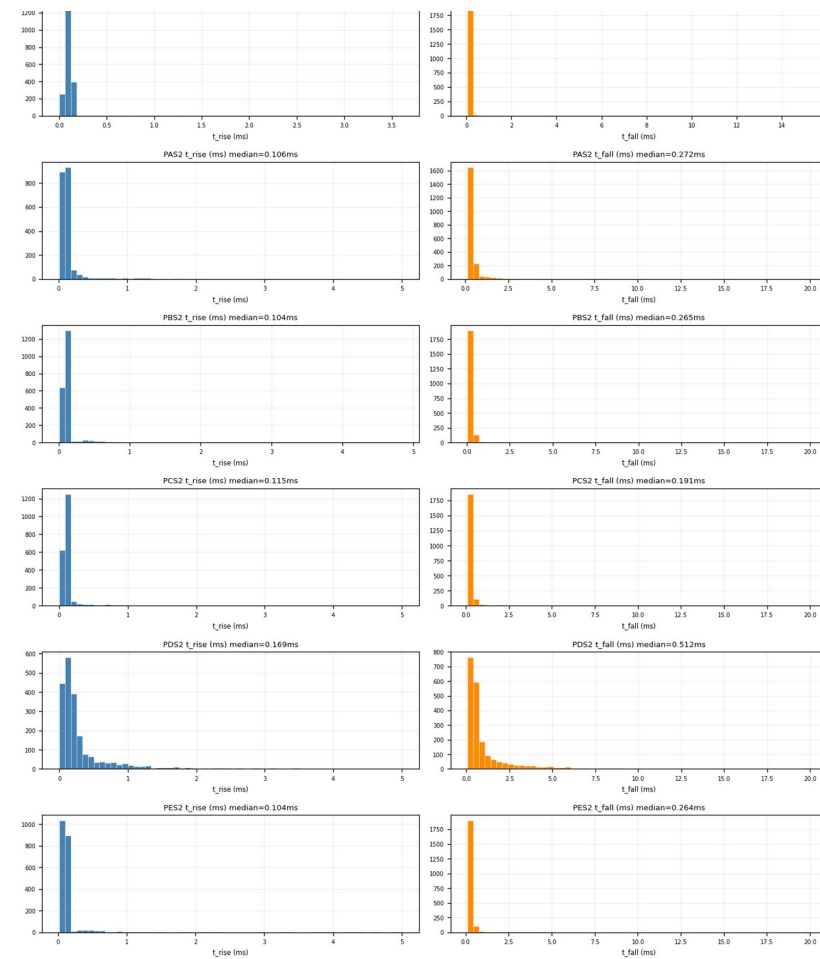
figure: ip_fit_align results — Zip7_time_constants.png

```
processing: raw MEDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 3000-10000) -> normalize by 0.08M trace peak (no peak window)
-> 2-imp fit y = A*exp(-t/tau_fall) - exp(-t/tau_rise); All 5 params free incl. pretrigger (source fit, pretrigger = 10000-3000, maxlev=10000)
-> fit ok => amp0 and det (cutoff fall) -> NRMSE fit residual (fitted pulse peak, RECORDED ONLY (no cut))
-> align => shift the MEDAS LP trace by (fitted pretrigger - 10000), sub-sample to integer
this figure: fitted tau_rise / tau_fall distributions (ms) of fit_ok events; free-pretrigger fit; loose bounds tau_rise=5ms tau_fall=20ms
```

Zip7 — τ_{rise} / τ_{fall} (fit_ok events, free-pretrigger fit)



part 1/2 (top to bottom)

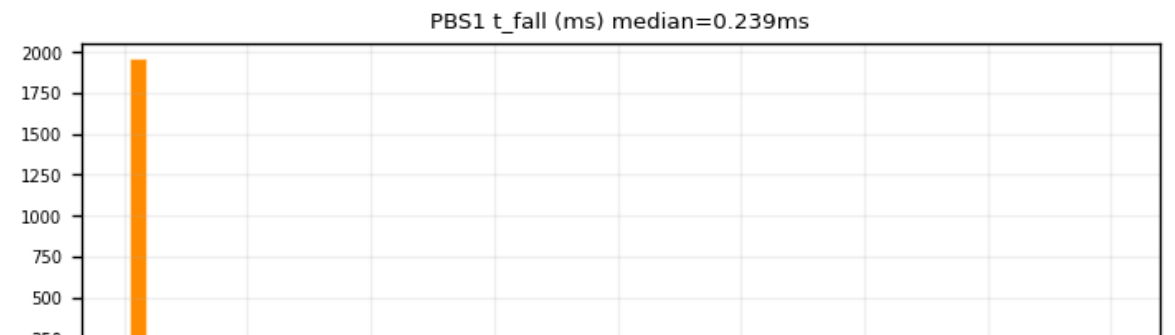
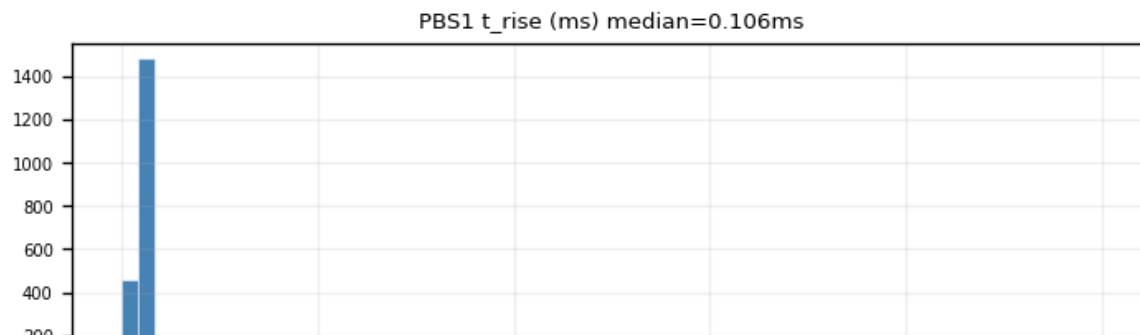
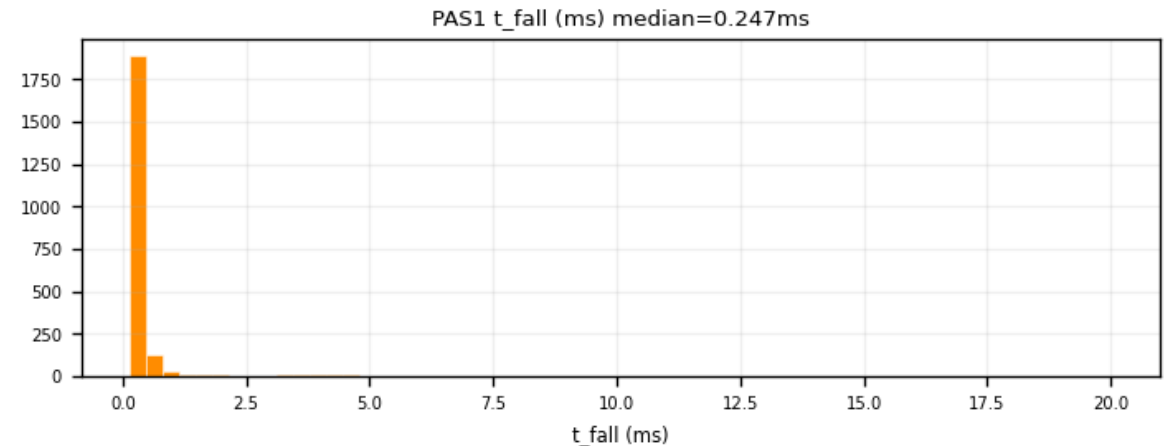
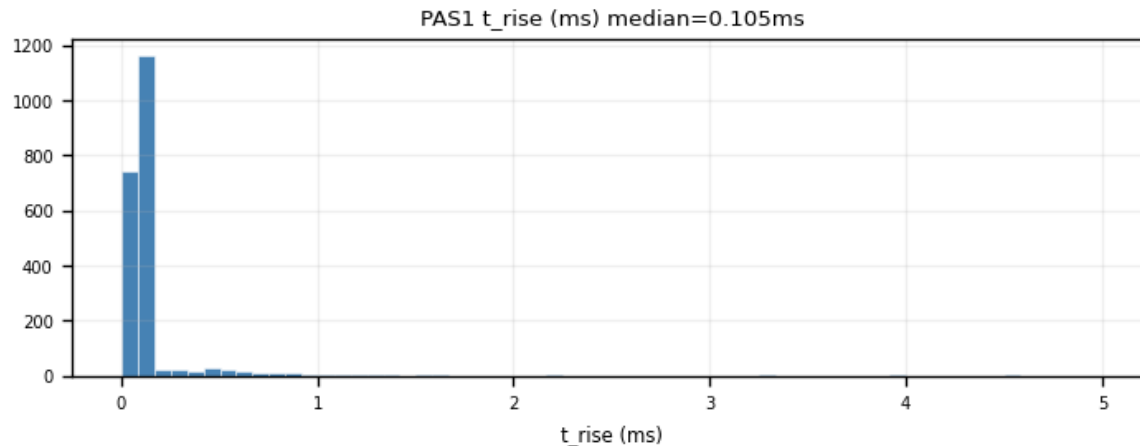


part 2/2 (top to bottom)

Backup gallery — Z7 · time_constants — fitted τ_{rise} / τ_{fall} distributions (detail, 1:1)

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and  $0 < t_{\text{rise}} < t_{\text{fall}}$  -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: fitted  $t_{\text{rise}}$  /  $t_{\text{fall}}$  distributions (ms) of fit_ok events; free-pretrigger fit, loose bounds  $t_{\text{rise}} < 5\text{ms}$   $t_{\text{fall}} < 20\text{ms}$ 
```

Zip7 — t_{rise} / t_{fall} (fit_ok events, free-pretrigger fit)



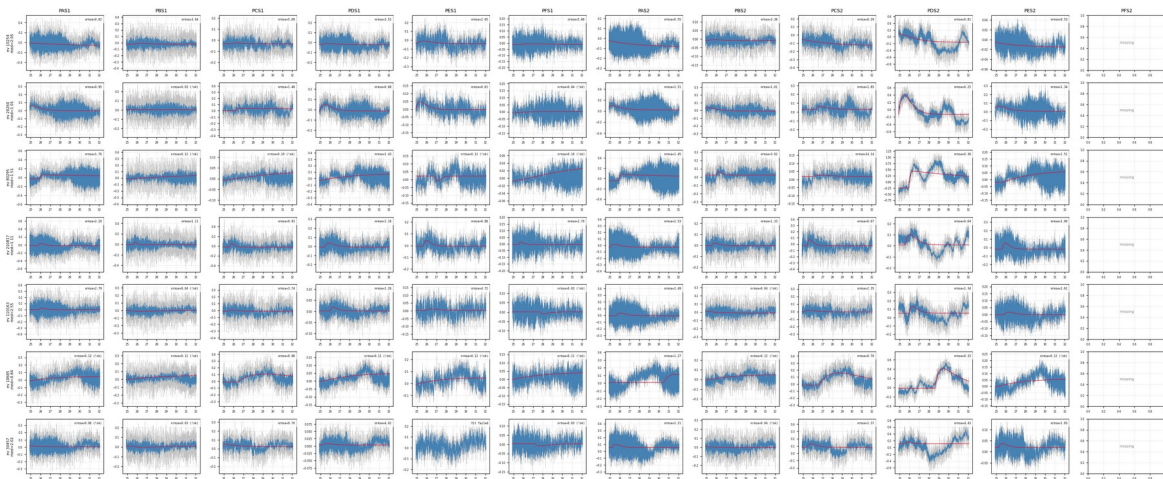
Backup gallery — Z7 · slow_rise_events — the NRMSE-rejected population

Events whose median NRMSE across channels > 0.4 . Raw traces show no pulse in any channel: the rejected population is noise triggers, not physics.

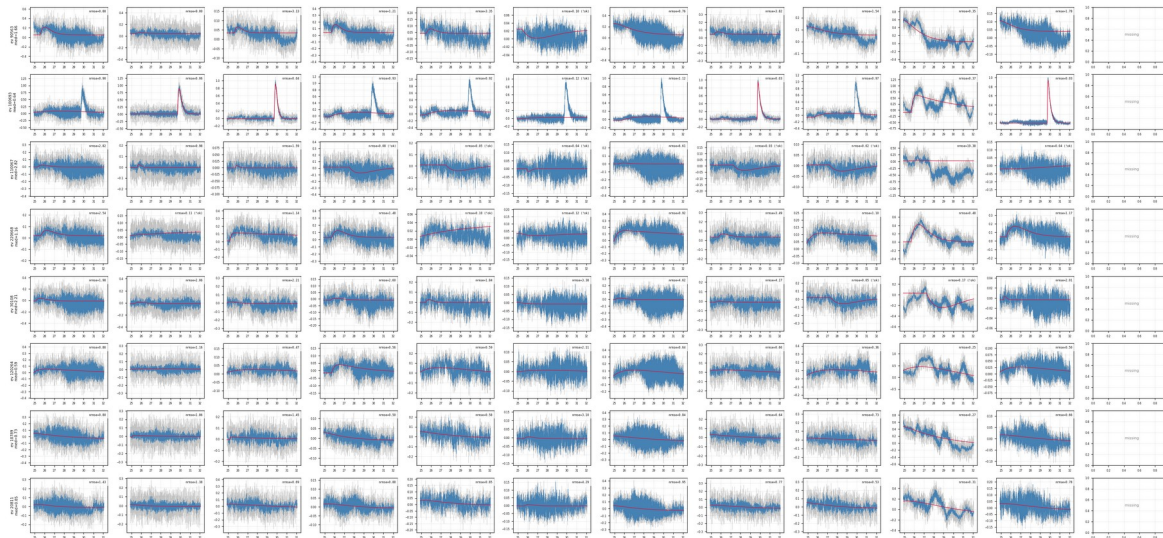
figure: ip_fit_align results — zip7_slow_rise_events.png

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Zip7 — NRMSE-rejected (red) population, first 15 events with median NRMSE > 0.4 : raw (gray) vs LP (blue) vs 2-exp fit (red)



part 1/2 (top to bottom)

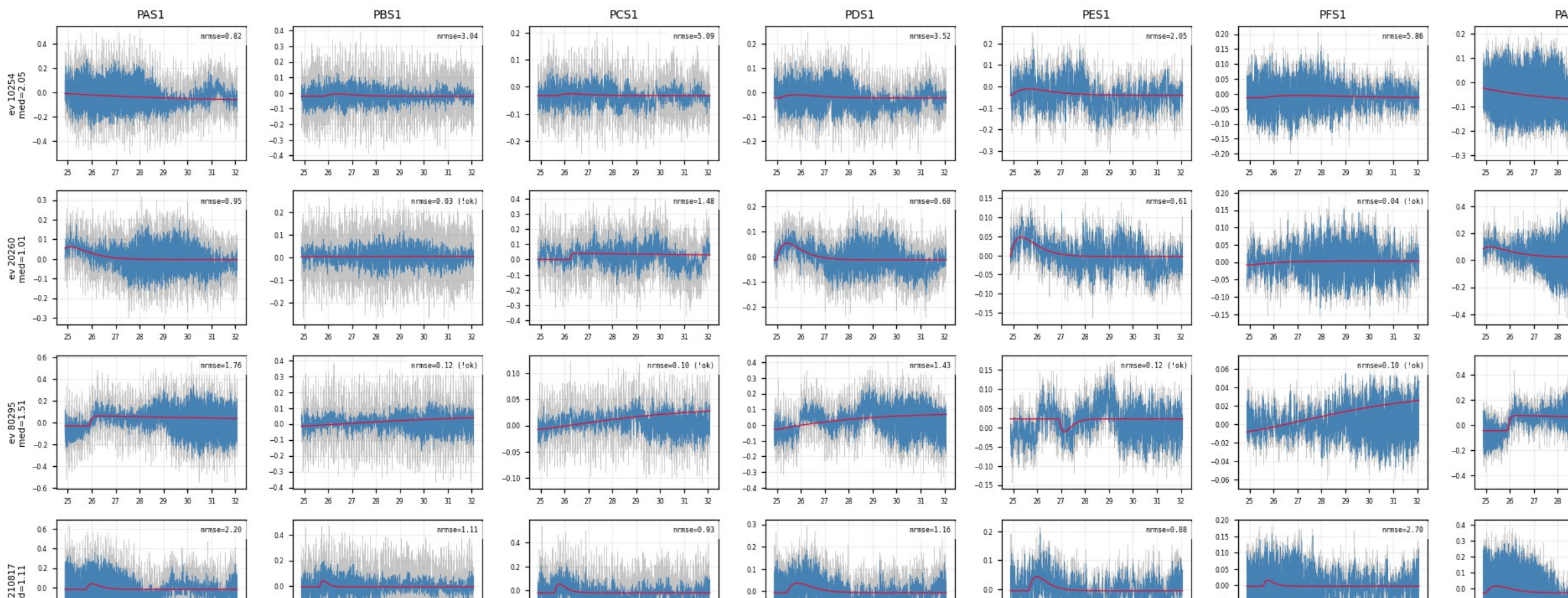


part 2/2 (top to bottom)

Backup gallery — Z7 · slow_rise_events — the NRMSE-rejected population (detail, 1:1)

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
 -> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
 -> 2-exp fit $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$, ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050+3000, maxfev=10000)
 -> fit ok := amp>0 and 0<t_rise< fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
 -> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
 this figure: slow-onset candidates = the RED (rejected) population of overlay fan cut: events whose MEDIAN NRMSE across fit ok channels exceeds 0.4 (first 15 in storage order); one ROW per event, one COLUMN per channel, gray = raw, blue = 100kHz LP; red = free-pretrigger 2-exp fit; judge from the raw traces whether these are noise triggers or genuinely slow pulses

Zip7 — NRMSE-rejected (red) population, first 15 events with median NRMSE >



Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} \geq 0.2$ ms: genuinely slow, well-fit pulses — the faint displaced bundle seen in aligned overlays; kept, and exactly what NxM multi-templates are for.

Median NRMSE ≤ 0.4 AND median $\tau_{\text{rise}} \geq 0.2$ ms: genuinely slow, well-fit pulses — the faint displaced bundle seen in aligned overlays; kept, and exactly what NxM multi-templates are for.

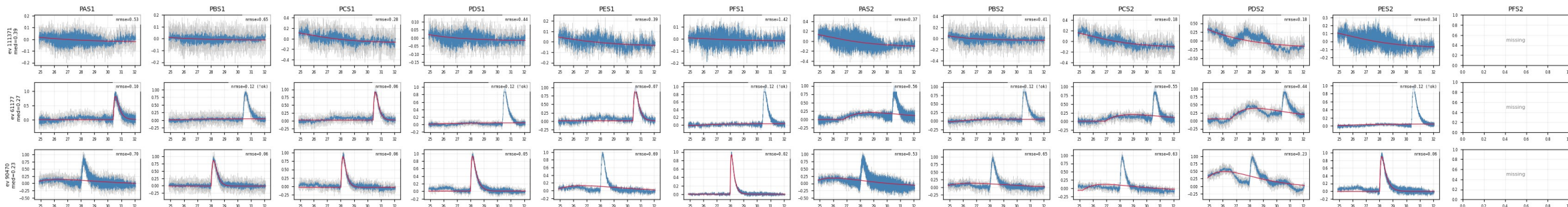
```

processing: read MIDAS trace > 100 kHhz 4th-order Butterworth (low-pass) (systo fixstep, steady-state)
> baseline subtract (median of samples 2000-12000) > normalize by GLOBAL trace peak (no peak window)
> 2-exp fit = A*(exp(-t/tau_fit) - exp(-t/tau_rise)), ALL 5 params free int., pretrigcor (curv fit, pretrigcor in 16050-3000, maxtwe=10000)
> maxtwe = 10000 and doct rise=fall > NOISE (GHOST) ONLY (no cut)
> align -> shift the MEASURED Up/Flt trace by (fitted pretrigcor - 16050), sub-sample nq.inter

```

this figure: GHOST population of aligned overlap = genuinely slow, WELL-FITs, median NOISE across fit pk channels ~ 0.4 and median SNR ~ 1.5. It is strictly the 16050 to 3000 range, and the 16050 to 3000 range is the only range where the overlap is displaced below in aligned overlap: one ROW per event, one COLUMN per channel, $\text{gray} = \text{raw}$, $\text{blue} = 100\text{MHz}$, $\text{red} = 2\text{-exp}$ fit

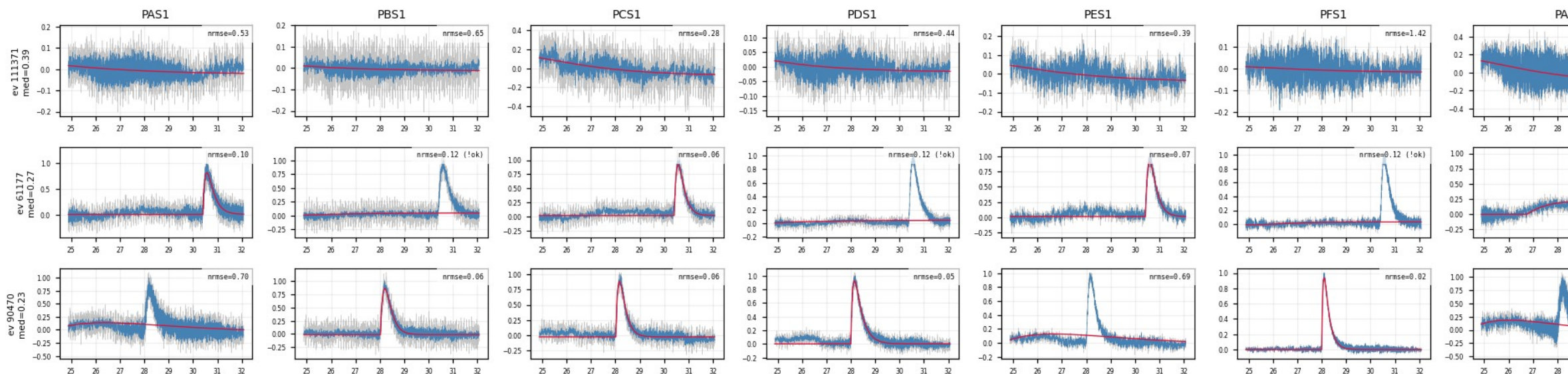
Zip7 — ghost population (well-fit slow-rise events), first 3 events with median NRMSE ≤ 0.4 and median t rise > 0.20 ms: raw (gray) vs LP (blue) vs 2-exp fit (red)



Backup gallery — Z7 · shadow_events — well-fit slow-rise (echo-trigger) events (detail, 1:1)

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$, ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxfev=10000)
-> fit ok := amp>0 and 0<t_rise< fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: GHOST population of aligned overlay = genuinely slow, WELL-FIT events: median NRMSE across fit ok channels <= 0.4 AND median t_rise > 0.20 ms (first 3 in storage order); onset aligns at 16050 but the peak comes later, producing the faint displaced bundle in aligned_overlay; one ROW per event, one COLUMN per channel, gray = raw, blue = 100kHz LP, red = 2-exp fit

Zip7 — ghost population (well-fit slow-rise events), first 3 events with median NRMSE <= 0.4 and n



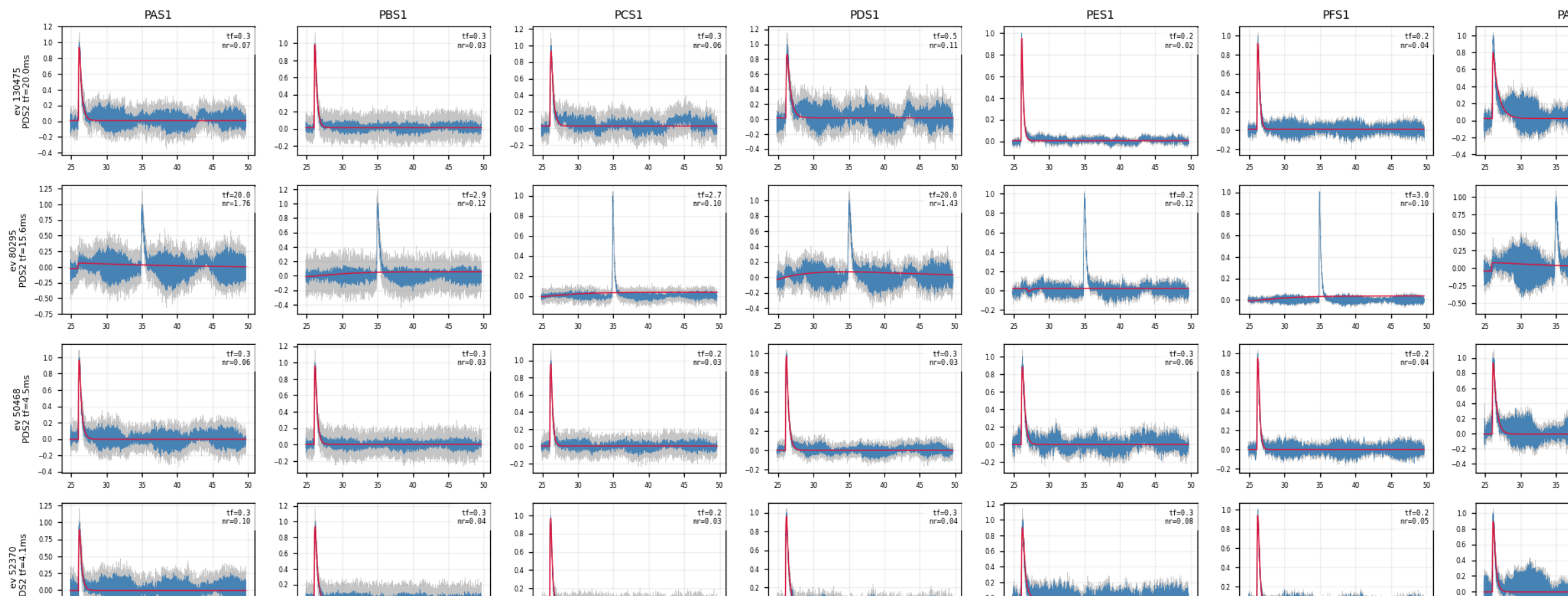
~~reference)~~

10 random events with $PDS2_{\tau_{\text{fall}}} > 1.5$ ms drawn in every channel figure. 10 fit-align results — zip7_PDS2_slow_fall.png
physics, no event-level cut.

Backup gallery — Z7 · slow_fall_events — long- τ _fall study (PDS2 reference) (detail, 1:1)

processing: raw MIDAS trace -> 100 KHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}}))$, ALL 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxfev=10000)
-> fit ok := amp>0 and 0<=rise<=fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: random 10 events PDS2 $t_{\text{fall}} > 1.5\text{ms}$ (fit_ok NRMSE<=0.4); row=event col=channel; gray=raw blue=LP red=fit

Zip7 10 random events PDS2 $t_{\text{fall}} > 1.5\text{ms}$: raw(gray)



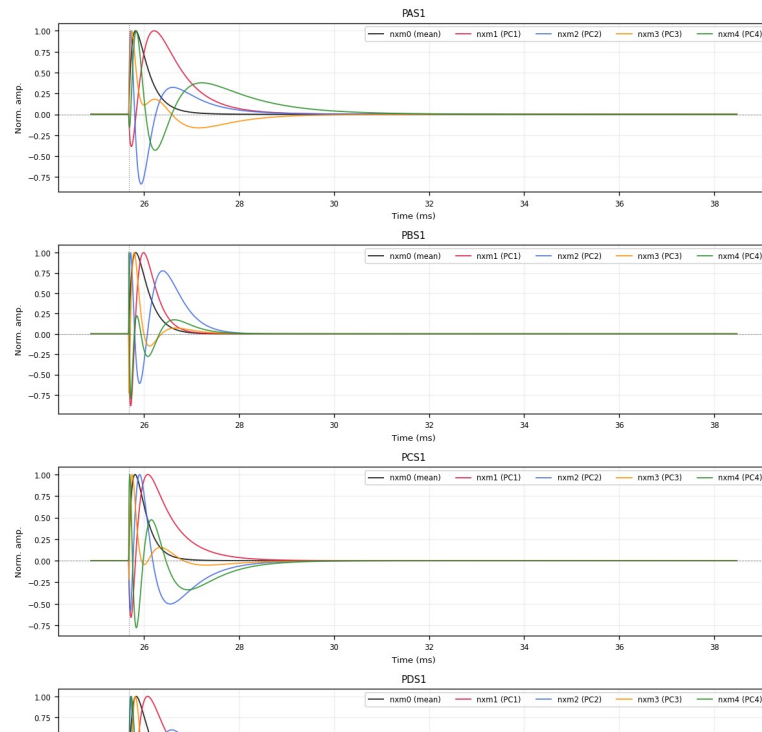
Backup gallery — Z7 · deliverables/nxm — PCA templates nxm0–4

figure: lp_fit_align results — zip7_pca_templates.png

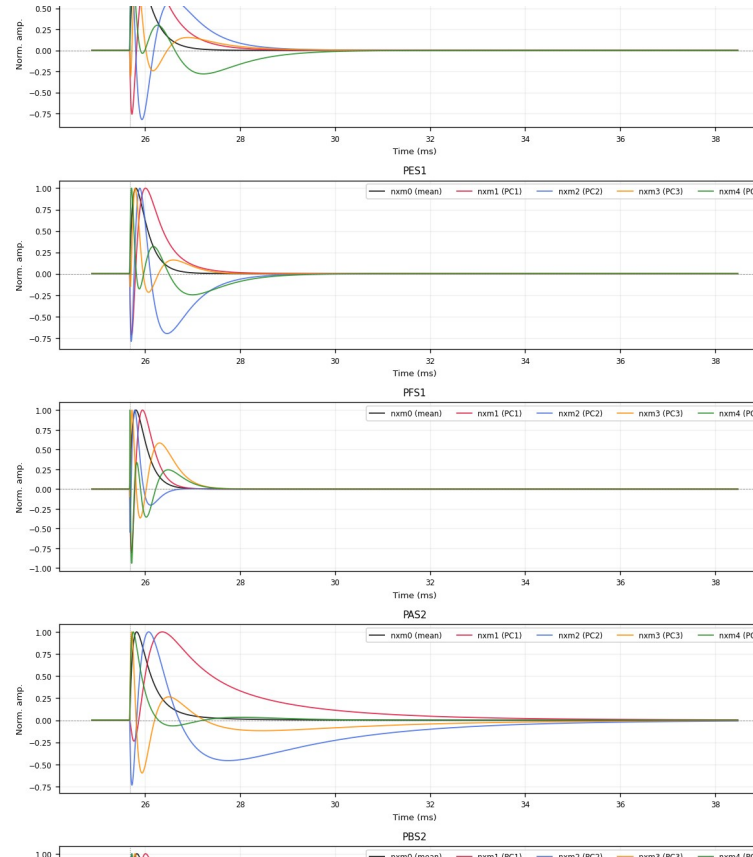
nxm0 = mean curve; nxm1–4 = PCA components (oscillating basis vectors, may be negative), all peak-normalized to 1; PC1+PC2 capture 96–98% of the shape variance.

```
processing: raw NIDA5 trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-seg fit  $y = A(\exp(-t/\tau_{\text{fall}}) - \exp(-t/\tau_{\text{rise}}))$ . All 5 params free incl. pretrigger (curve fit, pretrigger in 16050-3000, maxfev=10000)
fit ok => amp and offset fall -> NDRSE -> NML fit residuals (fitted distal peak, RECORD0 ONLY (no cut))
-> align => shift the RECORDED LP trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: NxM PCA templates. All peak-normalized to unit peak abs (root mean + root 4 PCA components 1-4). Population = fitted
2-exp curves with fit_ok AND NDRSE<0.4 AND t_rise=0.30ms at common pretrigger 16050; per-channel PCA. Real pulse = sum_i amp_i
* nxm_i.
```

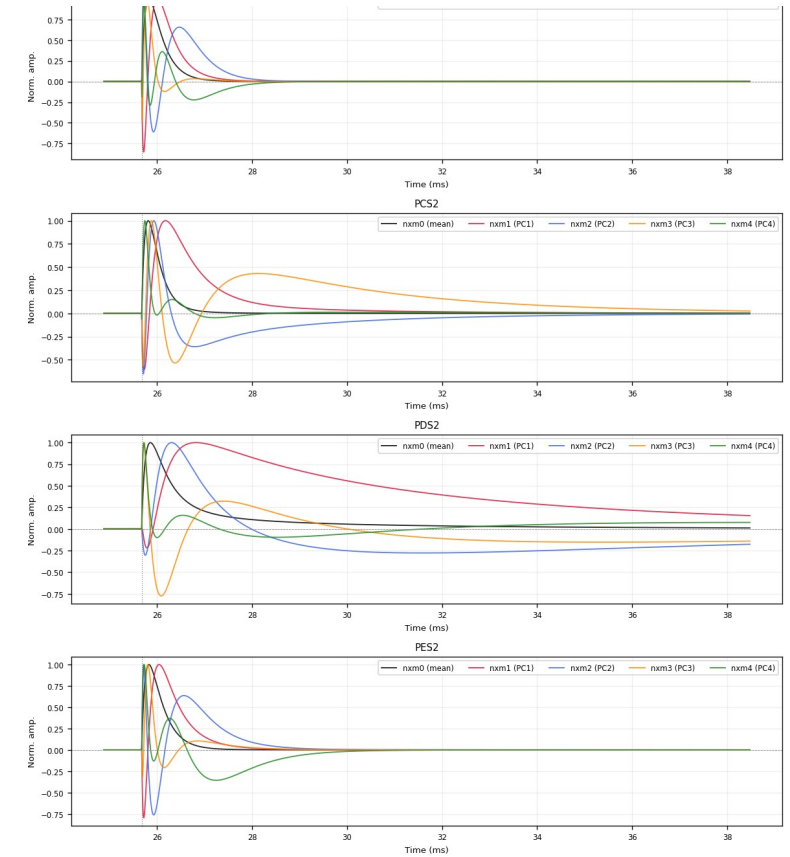
Zip7 - NxM PCA templates (all peak-normalized to 1)



part 1/3 (top to bottom)



part 2/3 (top to bottom)

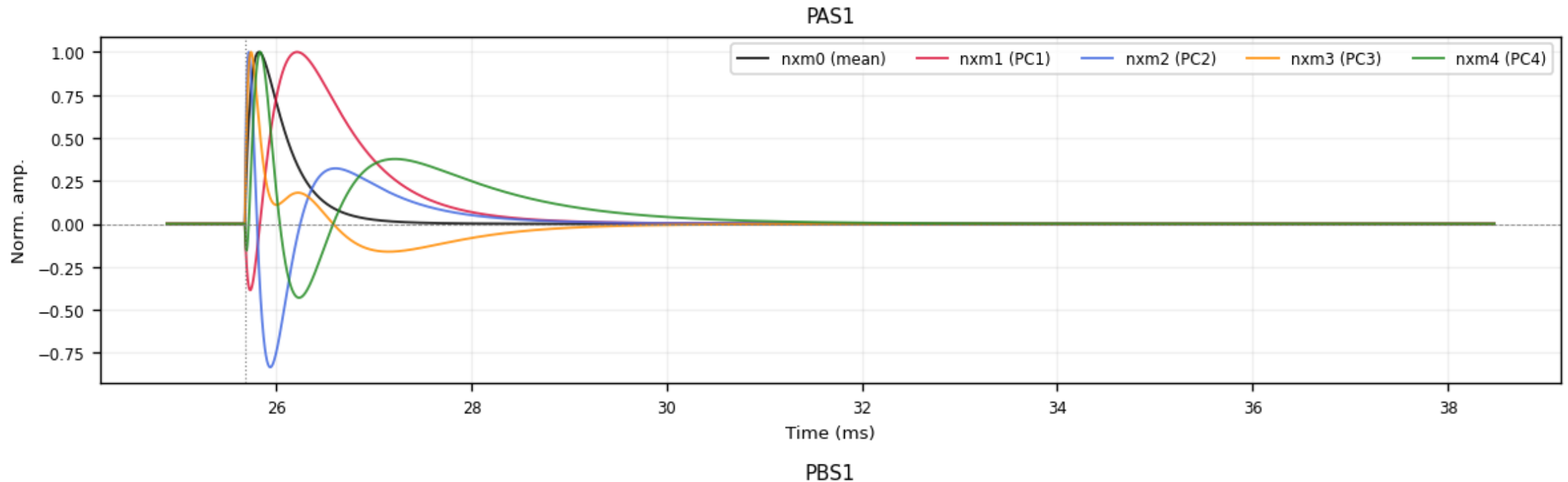


part 3/3 (top to bottom)

Backup gallery — Z7 · deliverables/nxm — PCA templates nxm0–4 (detail, 1:1)

```
processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit  $y = A(\exp(-t/t_{\text{fall}}) - \exp(-t/t_{\text{rise}}))$ , ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050+-3000, maxfev=10000)
-> fit_ok := amp>0 and 0<t_rise<t_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP_trace by (fitted pretrigger - 16050), sub-sample np.interp
this figure: NxM PCA templates, ALL peak-normalized to unit peak-abs (nxm0 mean + nxm1-4 PCA components 1-4). Population = fitted
2-exp curves with fit_ok AND NRMSE<=0.4 AND t_rise<=0.30ms at common pretrigger 16050; per-channel PCA. Real pulse = sum_i amp_i
* nxm_i.
```

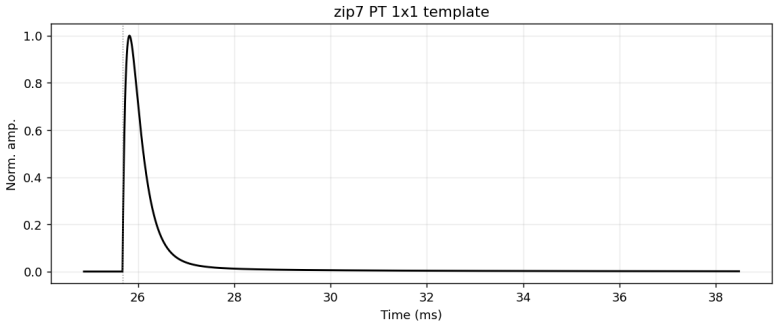
Zip7 - NxM PCA templates (all peak-normalized to 1)



Backup gallery — Z7 · deliverables/1x1 — summed templates PT / PS1 / PS2

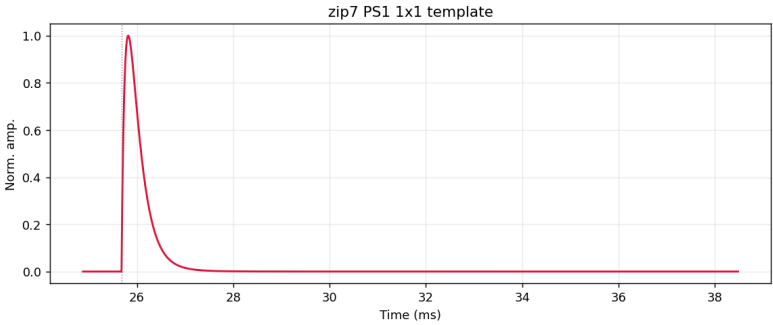
Single template (1x1) summed curves; peak-normalized average of the per-channel nxm0 templates — PT = all channels, PS1 / PS2 = side-1 / side-2 only.

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/t_{fall}) - \exp(-t/t_{rise}))$, ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050-3000, maxfev=10000)
-> fit_ok := amp>0 and bct_risect_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample up.interp
this figure: PT 1x1 template of zip7: peak-normalized average of the per-channel nxm0 PCA templates (all channels); channel nxm0 = mean of the clean fitted-curve population (fit_ok, NRMSE<=0.4, t_rise<=0.3ms)



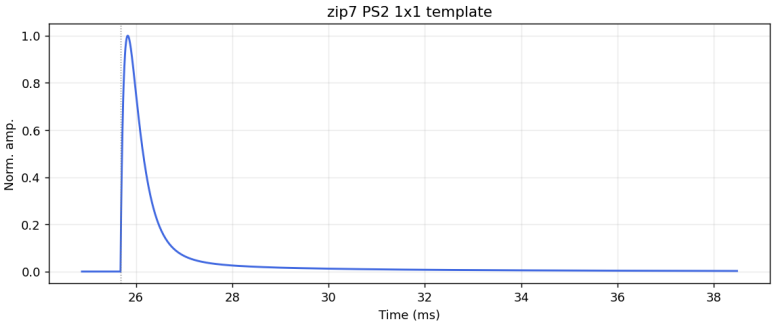
Z7 PT

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/t_{fall}) - \exp(-t/t_{rise}))$, ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050-3000, maxfev=10000)
-> fit_ok := amp>0 and bct_risect_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample up.interp
this figure: PS1 1x1 template of zip7: peak-normalized average of the per-channel nxm0 PCA templates (side-1 channels); channel nxm0 = mean of the clean fitted-curve population (fit_ok, NRMSE<=0.4, t_rise<=0.3ms)



Z7 PS1

processing: raw MIDAS trace -> 100 kHz 4th-order Butterworth low-pass (scipy sosfilt, steady-state init)
-> baseline subtract (median of samples 2000-12000) -> normalize by GLOBAL trace peak (no peak window)
-> 2-exp fit $y = A(\exp(-t/t_{fall}) - \exp(-t/t_{rise}))$, ALL 5 params free incl. pretrigger (curve_fit, pretrigger in 16050-3000, maxfev=10000)
-> fit_ok := amp>0 and bct_risect_fall -> NRMSE := RMS(fit residual)/fitted pulse peak, RECORDED ONLY (no cut)
-> align := shift the MEASURED LP trace by (fitted pretrigger - 16050), sub-sample up.interp
this figure: PS2 1x1 template of zip7: peak-normalized average of the per-channel nxm0 PCA templates (side-2 channels); channel nxm0 = mean of the clean fitted-curve population (fit_ok, NRMSE<=0.4, t_rise<=0.3ms)



Z7 PS2